



On the (un)decidability of fuzzy description logics under Łukasiewicz t -norm

Marco Cerami^a, Umberto Straccia^{b,*}

^a IIIA, CSIC, Bellaterra, Catalonia, Spain

^b ISTI, CNR, Pisa, Italy

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ABSTRACT

Recently there have been some unexpected results concerning Fuzzy Description Logics (FDLs) with General Concept Inclusions (GCI). They show that, unlike the classical case, the DL $\mathcal{ALC}$ with GCI does not have the finite model property under Łukasiewicz Logic or Product Logic, the proposed reasoning algorithms are neither correct nor complete and, specifically, knowledge base satisfiability is an undecidable problem for Product Logic.

In this work, we show that knowledge base satisfiability is also an undecidable problem for Łukasiewicz Logic. We additionally provide a decision algorithm for acyclic $\mathcal{ALC}$ knowledge bases under Łukasiewicz Logic via a Mixed Integer Linear Programming (MILP) based procedure (note, however, that the decidability of this problem is already known). While similar MILP based algorithms have been proposed in the literature for acyclic $\mathcal{ALC}$ knowledge bases under Łukasiewicz Logic, none of them exhibit formal proofs of their correctness and completeness, which is the additional contribution here.

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1. Introduction

Description Logics (DLs) [3] play a key role in the design of *Ontologies*. Indeed, DLs are important as they are essentially the theoretical counterparts of the *Web Ontology Language OWL 2* [41], the standard language to represent ontologies.

It is very natural to extend DLs to the fuzzy case and several fuzzy extensions of DLs can be found in the literature. For a recent survey on the advances in the field of fuzzy DLs, we refer the reader to [40]. Besides the enrichment of DLs with fuzzy features, one of the challenges of the research in this community is the fact that different families of fuzzy operators (or fuzzy logics) lead to fuzzy DLs with different computational properties.

Decidability of fuzzy DLs is often shown by adapting crisp DL tableau-based algorithms to the fuzzy DL case [15,44–46,48,52], or a reduction to classical DLs [9,10,12,18,47], or relying on some Mathematical Fuzzy Logic [29] based procedures [23,24,30,31].

However, recently there have been some unexpected surprises [5–8]. Indeed, unlike the classical case, for the DL $\mathcal{ALC}$ with GCI (i) [8] shows that it does not have the finite model property under Łukasiewicz Logic or Product Logic, illustrates that some algorithms are neither complete nor correct, and shows some interesting conditions under which decidability is still guaranteed; and (ii) [6,7] show that knowledge base satisfiability is an undecidable problem for it under Product Logic. Also worth mentioning is [21], which illustrates the undecidability of knowledge base satisfiability if one replaces the truth set $[0, 1]$ with complete De Morgan lattices equipped with a t -norm operator.

* Corresponding author. Address: Istituto di Scienza e Tecnologie dell'Informazione, Consiglio Nazionale delle Ricerche, Via G. Moruzzi, 1, 56124 Pisa (PI), Italy.

E-mail addresses: cerami@iia.csic.es (M. Cerami), straccia@isti.cnr.it (U. Straccia).

In this paper, we show that knowledge base satisfiability is an undecidable problem for the DL $\mathcal{ALC}$ with GCIs under $[0, 1]$ -valued Łukasiewicz Logic as well. We prove our results following conceptually the methods devised in [6,7,21].

Additionally, from a practical/implementation perspective, we provide a decision algorithm for acyclic $\mathcal{ALC}$ knowledge bases under Łukasiewicz Logic via a *Mixed Integer Linear Programming* (MILP) based procedure, which is a slightly variant of and conceptually identical to the ones described in previous works such as [12–15,48,52]. Please note, however, that the problem is already known to be decidable [8], but as pointed out in [8], besides being the results in [12–15,52] not completely correct, complete proofs have not been published so far, which is the additional contribution here. We point also out that the MILP based procedure is especially important as by means of it many other constructs can be implemented [11,13,16,17,48,50,51] and it is also the core of the *fuzzyDL* reasoner [11].

We next introduce briefly our fuzzy DL, then we illustrate the undecidability and decidability results, and conclude.

2. The FDL $\mathbb{L}\text{-}\mathcal{ALC}$

In this section we are going to introduce the general definitions of $\mathbb{L}\text{-}\mathcal{ALC}$ based on Łukasiewicz t -norm.

Syntax Let $\mathbf{A}$ be a set of *concept names*, $\mathbf{R}$ be a set of *role names*. Concept names denote unary predicates, while role names denote binary predicates. The set of $\mathbb{L}\text{-}\mathcal{ALC}$ concepts are built from concept names A (also called atomic concepts) using connectives and quantification constructs over roles R^1 as described by the following syntactic rules:

$$C \rightarrow \top \mid \perp \mid A \mid C_1 \sqcap C_2 \mid C_1 \sqcup C_2 \mid \neg C \mid \exists R.C \mid \forall R.C.$$

An *assertion axiom* is an expression of the form $\langle a; C, n \rangle$ (*concept assertion*, a is an instance of concept C to degree at least n)² or of the form $\langle (a_1, a_2); R, n \rangle$ (*role assertion* (a_1, a_2) is an instance of role R to degree at least n), where a, a_1, a_2 are individual names, C is a concept, R is a role name and $n \in (0, 1]$ is a rational (a truth value). An *ABox* $\mathcal{A}$ consists of a finite set of assertion axioms.

A *General Concept Inclusion* (GCI) axiom is of the form $\langle C_1 \sqsubseteq C_2, n \rangle$ (C_1 is a sub-concept of C_2 to degree at least n), where C_i is a concept and $n \in (0, 1]$ is a rational. A *Terminological Box* $\mathcal{T}$, also called *TBox*, is a finite set of GCIs. In what follows we will use the following short hands:

- $C_1 \rightarrow C_2$ for $\neg C_1 \sqcup C_2$;
- $C_1 \leftrightarrow C_2$ for $(C_1 \rightarrow C_2) \sqcap (C_2 \rightarrow C_1)$;
- $\min\{C_1, C_2\}$ for $C_1 \sqcap (C_1 \rightarrow C_2)$, and $\min\{C_1, \dots, C_n\}$ for $\min\{\dots \min\{C_1, C_2\}, \dots\}$;
- $\max\{C_1, C_2\}$ for $(C_1 \rightarrow C_2) \rightarrow C_2$ and $\max\{C_1, \dots, C_n\}$ for $\max\{\dots \max\{C_1, C_2\}, \dots\}$;
- $n \cdot C$ for the n -ary disjunction $C \sqcup \dots \sqcup C$;
- $C_1 \sqsubseteq C_2$ for $\langle C_1 \sqsubseteq C_2, 1 \rangle$ and $a:C$ for $\langle a; C, 1 \rangle$;
- $C_1 \equiv C_2$ for the two axioms $C_1 \sqsubseteq C_2$ and $C_2 \sqsubseteq C_1$ (or, equivalently for axiom $\top \sqsubseteq C_1 \leftrightarrow C_2$).

Finally, a *knowledge base* $\mathcal{K} = \langle \mathcal{T}, \mathcal{A} \rangle$ consists of a TBox $\mathcal{T}$ and an ABox $\mathcal{A}$. With $\text{sub}(\mathcal{K})$ we denote the set of (sub) concept expressions occurring in $\mathcal{K}$.

Semantics. From a semantics point of view, an axiom $\langle \alpha, n \rangle$ constrains the truth degree of the expression α to be at least n . In the following, we use $\otimes, \oplus, \ominus$ and $\Rightarrow$ to denote Łukasiewicz t -norm, t -conorm, negation function, and implication function, respectively [39]. They are defined as operations in $[0, 1]$ by means of the following functions:

$$\begin{aligned} n \otimes m &= \max\{0, n + m - 1\}, \\ n \oplus m &= \min\{1, n + m\}, \\ \ominus n &= 1 - n, \\ n \Rightarrow m &= \min\{1, 1 - n + m\}, \end{aligned}$$

where n and m are arbitrary elements in $[0, 1]$. As in the classical framework, the implication can be defined in terms of disjunction (whose semantics is the t -conorm) and negation in the usual way: $n \Rightarrow m = \ominus n \oplus m$. Note also that for any implication defined as the residuum of a continuous t -norm $\otimes$, i.e.,

$$n \Rightarrow m = \max\{r \mid n \otimes r \leq m\},$$

we have that this is equivalent to the condition

$$m \geq n \otimes r \text{ iff } (n \Rightarrow m) \geq r. \quad (1)$$

We will use $n \Leftrightarrow m$ as shorthand for $(n \Rightarrow m) \otimes (m \Rightarrow n)$.

A *fuzzy interpretation* is a pair $\mathcal{I} = (\mathcal{A}^{\mathcal{I}}, \mathcal{I}^{\mathcal{I}})$ consisting of a non-empty (crisp) set $\mathcal{A}^{\mathcal{I}}$ (the *domain*) and of a *fuzzy interpretation function* $\mathcal{I}^{\mathcal{I}}$ that assigns:

¹ Each symbol may have super- and/or subscripts.

² Often, in fuzzy DLs one may encounter concept assertions of the form $\langle a; C \geq n \rangle$ and $\langle a; C \leq n \rangle$ instead. Note that the latter is equivalent to $\langle a; \neg C, 1 - n \rangle$.

Table 1
Semantics for $\mathbb{L}\text{-}\mathcal{ALC}$.

$\perp^{\mathcal{I}}(x) = 0$
$\top^{\mathcal{I}}(x) = 1$
$(C \sqcap D)^{\mathcal{I}}(x) = C^{\mathcal{I}}(x) \otimes D^{\mathcal{I}}(x)$
$(C \sqcup D)^{\mathcal{I}}(x) = C^{\mathcal{I}}(x) \oplus D^{\mathcal{I}}(x)$
$(\neg C)^{\mathcal{I}}(x) = \ominus C^{\mathcal{I}}(x)$
$(\forall R.C)^{\mathcal{I}}(x) = \inf_{y \in \Delta^{\mathcal{I}}} \{R^{\mathcal{I}}(x, y) \Rightarrow C^{\mathcal{I}}(y)\}$
$(\exists R.C)^{\mathcal{I}}(x) = \sup_{y \in \Delta^{\mathcal{I}}} \{R^{\mathcal{I}}(x, y) \otimes C^{\mathcal{I}}(y)\}$

1. To each atomic concept A a function $A^{\mathcal{I}} : \Delta^{\mathcal{I}} \rightarrow [0, 1]$,
2. To each role R a function $R^{\mathcal{I}} : \Delta^{\mathcal{I}} \times \Delta^{\mathcal{I}} \rightarrow [0, 1]$,
3. To each individual a an element $a^{\mathcal{I}} \in \Delta^{\mathcal{I}}$ such that $a^{\mathcal{I}} \neq b^{\mathcal{I}}$ if $a \neq b$ (*Unique Name Assumption*, different individuals denote different objects of the domain).

The fuzzy interpretation function is extended to complex concepts as specified in Table 1 (where $x, y \in \Delta^{\mathcal{I}}$ are elements of the domain). Hence, for every complex concept C we get a function $C^{\mathcal{I}} : \Delta^{\mathcal{I}} \rightarrow [0, 1]$. The *satisfiability of axioms* is then defined by the following conditions:

1. $\mathcal{I}$ satisfies an axiom $\langle a; C, n \rangle$ if $C^{\mathcal{I}}(a^{\mathcal{I}}) \geq n$,
2. $\mathcal{I}$ satisfies an axiom $\langle (a, b); R, n \rangle$ if $R^{\mathcal{I}}(a^{\mathcal{I}}, b^{\mathcal{I}}) \geq n$,
3. $\mathcal{I}$ satisfies an axiom $\langle C \sqsubseteq D, n \rangle$ if $(C \sqsubseteq D)^{\mathcal{I}} \geq n$, where

$$(C \sqsubseteq D)^{\mathcal{I}} = \inf_{x \in \Delta^{\mathcal{I}}} \{C^{\mathcal{I}}(x) \Rightarrow D^{\mathcal{I}}(x)\}.$$

It is interesting to point out that the satisfaction of a GCI of the form $\langle C \sqsubseteq D, 1 \rangle$ is exactly the requirement that $\forall x \in \Delta^{\mathcal{I}}, C^{\mathcal{I}}(x) \leq D^{\mathcal{I}}(x)$ (i.e., Zadeh's set inclusion); hence, in this particular case for the satisfaction it only matters the partial order and not the exact value of the implication $\Rightarrow$.

As usual we will say that a fuzzy interpretation $\mathcal{I}$ *satisfies* (is a model of) a KB $\mathcal{K}$ in case that it satisfies all axioms in $\mathcal{K}$. And it is said that a fuzzy KB $\mathcal{K}$ is *satisfiable* (has a model) iff there exists a fuzzy interpretation $\mathcal{I}$ satisfying every axiom in $\mathcal{K}$. A fuzzy KB $\mathcal{K}$ *entails* an axiom $\langle \alpha, n \rangle$ (denoted $\mathcal{K} \models \langle \alpha, n \rangle$) iff any model of $\mathcal{K}$ also satisfies $\langle \alpha, n \rangle$.

Note that, as long as $\mathcal{K}$ is satisfiable, the problem of determining whether $\mathcal{K} \models \langle (a, b) : R, n \rangle$ can easily be determined by checking if there is $\langle (a, b) : R, m \rangle \in \mathcal{A}$ with $m \geq n$. Suppose, in fact, that $\mathcal{K}$ is a satisfiable knowledge base with $\langle (a, b) : R, m \rangle \notin \mathcal{A}$ for every $m \geq n$. Since $\mathcal{K}$ is satisfiable, then it has at least a model $\mathcal{I}$. Without loss of generality, we can suppose that $R^{\mathcal{I}}(a^{\mathcal{I}}, b^{\mathcal{I}}) \geq n$, otherwise $\mathcal{K} \models \langle (a, b) : R, n \rangle$ holds trivially. From $\mathcal{I}$ we can easily build a model $\mathcal{I}'$ where:

1. $\Delta^{\mathcal{I}'} = \Delta^{\mathcal{I}} \cup \{z\}$, with $z \notin \Delta^{\mathcal{I}}$ and $z \neq a^{\mathcal{I}}$, for every individual name a appearing in $\mathcal{K}$,
2. $R^{\mathcal{I}'}(z, x) = R^{\mathcal{I}}(b^{\mathcal{I}}, x)$, for every $x \in \Delta^{\mathcal{I}}$,
3. $R^{\mathcal{I}'}(x, z) = R^{\mathcal{I}}(x, b^{\mathcal{I}})$, for every $x \in \Delta^{\mathcal{I}}$,
4. $R^{\mathcal{I}'}(b^{\mathcal{I}'}, x) = R^{\mathcal{I}}(b^{\mathcal{I}}, x)$, for every $x \in \Delta^{\mathcal{I}}$,
5. $R^{\mathcal{I}'}(x, b^{\mathcal{I}'}) = R^{\mathcal{I}}(x, b^{\mathcal{I}})$, for every $x \in \Delta^{\mathcal{I}}$ such that $a^{\mathcal{I}} \neq x$,
6. $R^{\mathcal{I}'}(a^{\mathcal{I}'}, b^{\mathcal{I}'}) < n$.

Indeed, by items 1, 2 and 3, every assertion where b does not appear, keeps being satisfied. By items 4 and 5 every assertion where b appears keeps being satisfied as well (remember that, by assumption, $\langle (a, b) : R, m \rangle \notin \mathcal{A}$ for every $m \geq n$). Since $\mathcal{I}$ satisfies $\mathcal{T}$, then by items 1–5, we have that $\mathcal{I}'$ keeps satisfying $\mathcal{T}$. Finally, item 6 does not contradict any axiom in $\mathcal{K}$ but assertion $\langle (a, b); R, n \rangle$. Hence $\mathcal{I}'$ is a model of $\mathcal{K}$, but not of $\langle (a, b); R, n \rangle$. This proves that $\mathcal{K}$ does not entail $\langle (a, b); R, n \rangle$.

The *Best Entailment Degree* (BED) of an axiom α w.r.t. $\mathcal{K}$ [11,46] is defined as

$$bed(\mathcal{K}, \alpha) = \sup\{n | \mathcal{K} \models \langle \alpha, n \rangle\},$$

while the *Best Satisfiability Degree* (BSD) of a concept C w.r.t. $\mathcal{K}$ is defined as

$$bsd(\mathcal{K}, C) = \sup_{\mathcal{I} \models \mathcal{K}} \sup_{x \in \Delta^{\mathcal{I}}} C^{\mathcal{I}}(x).$$

Remark 1. Note that it is well known that (see [11]), given $\mathcal{K} = \langle \mathcal{T}, \mathcal{A} \rangle$, we can compute $bed(\mathcal{K}, a : C)$ as the minimal value of n such that $\langle \mathcal{T}, \mathcal{A} \cup \{ \langle a : \neg C, 1 - n \rangle \} \rangle$ is satisfiable. Informally, concerning $bed(\mathcal{K}, a : C)$ suppose the minimal value is $\bar{n}$. We

will know then that for any interpretation $\mathcal{I}$ satisfying the knowledge base such that $(a : C)^{\mathcal{I}} < \bar{n}$, the starting set is unsatisfiable and, thus, $(a : C)^{\mathcal{I}} \geq \bar{n}$ has to hold. Which means that $\text{bed}(\mathcal{K}, a : C) = \bar{n}$.

Similarly, for $C \sqsubseteq D$, we can compute $\text{bed}(\mathcal{K}, C \sqsubseteq D)$ as the minimal value of n such that $\mathcal{K} = \langle \mathcal{T}, \mathcal{A} \cup \{(a : C, n_1)\} \cup \{(a : D, n_2)\} \rangle$ is satisfiable under the constraints expressing that $n_1 \Rightarrow n_2 \leq n$, $n_1 \in [0, 1]$ and $n_2 \in [0, 1]$, where a is new individual.

Therefore, the BED problem can be reduced to minimal satisfiability problem of a KB. Finally, concerning the best satisfiability bound problem, $\text{bsd}(\mathcal{K}, C)$ is determined by the maximal value of n such that $\langle \mathcal{T}, \mathcal{A} \cup \{(a : C, n)\} \rangle$ is satisfiable.

An important note is that in this paper, we mainly focus on witnessed models. This notion, firstly introduced by Hájek in [30] in the framework of FDL, corresponds to the restriction to the FDL language of the notion of witnessed model in the first-order language (see [32]). Specifically, a fuzzy interpretation $\mathcal{I}$ is said to be *strongly witnessed* iff it holds that for every complex concepts C, D , every role R , and every $x \in \Delta^{\mathcal{I}}$ there is some

1. $y \in \Delta^{\mathcal{I}}$ such that $(\exists R.C)^{\mathcal{I}}(x) = R^{\mathcal{I}}(x, y) \otimes C^{\mathcal{I}}(y)$.
2. $y \in \Delta^{\mathcal{I}}$ such that $(\forall R.C)^{\mathcal{I}}(x) = R^{\mathcal{I}}(x, y) \Rightarrow C^{\mathcal{I}}(y)$.
3. $y \in \Delta^{\mathcal{I}}$ such that $(C \sqsubseteq D)^{\mathcal{I}} = C^{\mathcal{I}}(y) \Rightarrow D^{\mathcal{I}}(y)$.

If $\mathcal{I}$ satisfies only conditions 1. and 2. then $\mathcal{I}$ is said to be *witnessed*. If $\mathcal{I}$ satisfies only condition 1. then $\mathcal{I}$ is said to be *weakly witnessed*. Note that for Łukasiewicz logic, condition 1. and 2. are equivalent, so $\mathcal{I}$ is weakly witnessed iff $\mathcal{I}$ is witnessed. Throughout the paper we will rely on the notion of witnessed interpretation only, but keep in mind that the results apply, thus, to weakly interpretations as well. Note also that it is obvious that all finite fuzzy interpretations (this means that $\Delta^{\mathcal{I}}$ is a finite set) are indeed strongly witnessed but the opposite is not true. Moreover, we recall that for Łukasiewicz logic, it holds that a KB is satisfiable iff witnessed satisfiable, while in general this is not true (see, e.g., [8,30,32]).

Finally, note that, without loss of generality (w.l.o.g.) we may assume that GCI are of the form $\langle \top \sqsubseteq C, n \rangle$ as any GCI of the form $\langle C_1 \sqsubseteq C_2, n \rangle$ can be replaced with the GCI $\langle \top \sqsubseteq C_1 \rightarrow C_2, n \rangle$.

3. Undecidability of $\mathbb{L}$ -ALC with GCIs

Our proof consists of a reduction of the *reverse* of the *Post Correspondence Problem* (PCP) and follows conceptually the one in [6,7,21]. PCP is well known to be undecidable [42], so is the reverse PCP, as shown next.

Definition 1 (PCP). An instance φ of the *Post Correspondence Problem* (PCP) is defined in the following way: let $v_1, \dots, v_p$ and $w_1, \dots, w_p$ be two finite lists of words over an alphabet $\Sigma = \{1, \dots, s\}$. A *solution* to φ is a non-empty sequence $i_1, i_2, \dots, i_k$, with $1 \leq i_j \leq p$ such that $v_{i_1} v_{i_2} \dots v_{i_k} = w_{i_1} w_{i_2} \dots w_{i_k}$. Given φ , the decision problem then is to decide whether a solution to φ exists or not.

For the sake of our purpose, we will rely on a variant of the PCP, which we call *Reverse PCP* (RPCP). Essentially, words are concatenated from right to left rather than from left to right.

Definition 2 (RPCP). An instance φ of the *Reverse Post Correspondence Problem* (RPCP) is defined in the following way: let $v_1, \dots, v_p$ and $w_1, \dots, w_p$ be two finite lists of words over an alphabet $\Sigma = \{1, \dots, s\}$. A *solution* to φ is a non-empty sequence $i_1, i_2, \dots, i_k$, with $1 \leq i_j \leq p$ such that $v_{i_k} v_{i_{k-1}} \dots v_{i_1} = w_{i_k} w_{i_{k-1}} \dots w_{i_1}$. Given φ , the decision problem then is to decide whether a solution to φ exists or not.

For a word $\mu = i_1 i_2 \dots i_k \in \{1, \dots, p\}^*$ we will use v_μ, w_μ to denote the words $v_{i_k} v_{i_{k-1}} \dots v_{i_1}$ and $w_{i_k} w_{i_{k-1}} \dots w_{i_1}$. We denote the empty string as ϵ and define v_ϵ as ϵ . The alphabet Σ consists of the first s positive integers. We can thus view every word in Σ^* as a natural number represented in base $s + 1$ in which 0 never occurs. Using this intuition, we will use the number 0 to encode the empty word.

Now we show that the reduction from PCP to RPCP is a very simple matter and it can be done through the transformation of the instance lists to the lists of their palindromes defined as follows: let $\Sigma = \{1, \dots, s\}$ be an alphabet and $v = t_1 t_2 \dots t_{|v|}$ a word over Σ , with $t_i \in \Sigma$, for $1 \leq i \leq |v|$, then the *palindrome* of v is defined as $\text{pal}(v) = t_{|v|} t_{|v|-1} \dots t_1$.

Lemma 3. Let $v_1, \dots, v_p$ and $w_1, \dots, w_p$ be two finite lists of words over an alphabet $\Sigma = \{1, \dots, s\}$. For every non-empty sequence $i_1, i_2, \dots, i_k$ with $1 \leq i_j \leq p$ it holds that

$$\begin{aligned} v_{i_1} v_{i_2} \dots v_{i_k} &= w_{i_1} w_{i_2} \dots w_{i_k} \\ \text{iff} \\ \text{pal}(v_{i_k}) \text{pal}(v_{i_{k-1}}) \dots \text{pal}(v_{i_1}) &= \text{pal}(w_{i_k}) \text{pal}(w_{i_{k-1}}) \dots \text{pal}(w_{i_1}). \end{aligned}$$

Proof. First we prove by induction on k , that, for every sequence $v = v_{i_1} v_{i_2} \dots v_{i_k}$ of words over Σ , it holds that $\text{pal}(v) = \text{pal}(v_{i_k}) \text{pal}(v_{i_{k-1}}) \dots \text{pal}(v_{i_1})$.

- The case $k = 1$ is straightforward.

- Let $v = v_{i_1} v_{i_2} \dots v_{i_k}$ and suppose, by inductive hypothesis, that $\text{pal}(v_{i_1} v_{i_2} \dots v_{i_{k-1}}) = \text{pal}(v_{i_{k-1}}) \text{pal}(v_{i_{k-2}}) \dots \text{pal}(v_{i_1})$. It follows that $\text{pal}(v) = \text{pal}(v_{i_1} v_{i_2} \dots v_{i_{k-1}}, v_{i_k}) = \text{pal}(v_{i_k}) \text{pal}(v_{i_{k-1}}) \dots \text{pal}(v_{i_1})$.

Since the palindrome of a word is unique, we have that, if $v_{i_1} v_{i_2} \dots v_{i_k} = w_{i_1} w_{i_2} \dots w_{i_k}$, then $\text{pal}(v_{i_1} v_{i_2} \dots v_{i_k}) = \text{pal}(w_{i_1} w_{i_2} \dots w_{i_k})$ and, thus, $\text{pal}(v_{i_k}) \text{pal}(v_{i_{k-1}}) \dots \text{pal}(v_{i_1}) = \text{pal}(w_{i_k}) \text{pal}(w_{i_{k-1}}) \dots \text{pal}(w_{i_1})$. $\square$

Corollary 4. *RPCP is undecidable.*

Proof. The proof is based on the reduction of PCP to RPCP. For every instance $\varphi = (v_1, w_1), \dots, (v_p, w_p)$ of PCP, let f be the function

$$f(\varphi) = (\text{pal}(v_1), \text{pal}(w_1)), \dots, (\text{pal}(v_p), \text{pal}(w_p)).$$

Clearly f is a computable function. Moreover, φ has a solution if and only if there exists a non-empty sequence $i_1, i_2, \dots, i_k$, with $1 \leq i_j \leq p$ such that $v_{i_1} v_{i_2} \dots v_{i_k} = w_{i_1} w_{i_2} \dots w_{i_k}$, that is, by Lemma 3,

$$\text{pal}(v_{i_k}) \text{pal}(v_{i_{k-1}}) \dots \text{pal}(v_{i_1}) = \text{pal}(w_{i_k}) \text{pal}(w_{i_{k-1}}) \dots \text{pal}(w_{i_1})$$

i.e., $f(\varphi)$ has a solution. Therefore, $\varphi \in \text{PCP}$ has a solution if and only if $f(\varphi) \in \text{RPCP}$ has a solution. $\square$

Undecidability of general KB satisfiability. We show the undecidability by a reduction of RPCP to KB satisfiability problems. Specifically, given an instance φ of RPCP, we will construct a Knowledge Base $\mathcal{O}_\varphi$ that is satisfiable iff φ has no solution.

In order to do this we will encode words v from the alphabet Σ as rational numbers $0.v$ in $[0, 1]$ in base $s + 1$; the empty word will be encoded by the number 0.

Let us define the following TBoxes:

$$\mathcal{T} := \{V \equiv V_1 \sqcup V_2, W \equiv W_1 \sqcup W_2\}$$

and for $1 \leq i \leq p$

$$\begin{aligned} \mathcal{T}_\varphi^i := & \{ \top \sqsubseteq \exists R_i. \top, \quad V \sqsubseteq (s+1)^{|v_i|} \cdot \forall R_i. V_1, (s+1)^{|v_i|} \cdot \exists R_i. V_1 \sqsubseteq V, W \sqsubseteq (s+1)^{|w_i|} \cdot \forall R_i. W_1, (s+1)^{|w_i|} \cdot \exists R_i. W_1 \sqsubseteq W \\ & \langle \top \sqsubseteq \forall R_i. V_2, 0.v_i \rangle, \langle \top \sqsubseteq \forall R_i. \neg V_2, 1 - 0.v_i \rangle, \langle \top \sqsubseteq \forall R_i. W_2, 0.w_i \rangle, \langle \top \sqsubseteq \forall R_i. \neg W_2, 1 - 0.w_i \rangle, \\ & A \sqsubseteq (s+1)^{\max\{|v_i|, |w_i|\}} \cdot \forall R_i. A, (s+1)^{\max\{|v_i|, |w_i|\}} \cdot \exists R_i. A \sqsubseteq A \}. \end{aligned}$$

Now, let

$$\mathcal{T}_\varphi = \mathcal{T} \sqcup \bigcup_{i=1}^p \mathcal{T}_\varphi^i.$$

Further we define the ABox $\mathcal{A}$ as follows:

$$\mathcal{A} := \{a : \neg V, a : \neg W, \langle a : A, 0.01 \rangle, \langle a : \neg A, 0.99 \rangle\}.$$

Finally, we define

$$\mathcal{O}_\varphi := \langle \mathcal{T}_\varphi, \mathcal{A} \rangle.$$

We now define the interpretation

$$\mathcal{I}_\varphi := (\mathcal{A}^{\mathcal{I}_\varphi}, \cdot^{\mathcal{I}_\varphi})$$

as follows:

- $\mathcal{A}^{\mathcal{I}_\varphi} = \{1, \dots, p\}^*$
- $a^{\mathcal{I}_\varphi} = \epsilon$
- $V^{\mathcal{I}_\varphi}(\epsilon) = W^{\mathcal{I}_\varphi}(\epsilon) = 0, A^{\mathcal{I}_\varphi}(\epsilon) = 0.01$, and for $1 \leq i \leq 2, V_i^{\mathcal{I}_\varphi}(\epsilon) = W_i^{\mathcal{I}_\varphi}(\epsilon) = 0$
- for all $\mu, \mu' \in \mathcal{A}^{\mathcal{I}_\varphi}$ and $1 \leq i \leq p$

$$R_i^{\mathcal{I}_\varphi}(\mu, \mu') = \begin{cases} 1, & \text{if } \mu' = \mu i, \\ 0, & \text{otherwise,} \end{cases}$$

- for every $\mu \in \mathcal{A}^{\mathcal{I}_\varphi}$, where $\mu = i_1 i_2 \dots i_k \neq \epsilon$,
 - $V^{\mathcal{I}_\varphi}(\mu) = 0.v_\mu, W^{\mathcal{I}_\varphi}(\mu) = 0.w_\mu$,
 - $A^{\mathcal{I}_\varphi}(\mu) = 0.01 \cdot (s+1)^{-\sum_{j \in \{i_1, i_2, \dots, i_k\}} \max\{|v_j|, |w_j|\}}$,
 - $V_1^{\mathcal{I}_\varphi}(\mu) = 0.v_{\bar{\mu}} \cdot (s+1)^{-|v_{i_k}|}, W_1^{\mathcal{I}_\varphi}(\mu) = 0.w_{\bar{\mu}} \cdot (s+1)^{-|w_{i_k}|}$, where $\bar{\mu} = i_1 i_2 \dots i_{k-1}$ (last index i_k is dropped from μ , and we assume that $0.\epsilon$ is 0),
 - $V_2^{\mathcal{I}_\varphi}(\mu) = 0.v_{i_k}, W_2^{\mathcal{I}_\varphi}(\mu) = 0.w_{i_k}$.

It is easy to see that $\mathcal{I}_\varphi$ is a witnessed model of $\mathcal{O}_\varphi$ (note that e.g., $(\forall R_i. V_1)^{\mathcal{I}_\varphi}(\mu) = V_1^{\mathcal{I}_\varphi}(\mu i)$).³

³ However, $\mathcal{I}_\varphi$ is not a strongly witnessed model of $\mathcal{O}_\varphi$.

Moreover, as in [6] it is possible to prove that, for every witnessed model $\mathcal{I}$ of $\mathcal{O}_\varphi$, there is a mapping g from $\mathcal{I}_\varphi$ to $\mathcal{I}$.

Lemma 5. *Let $\mathcal{I}$ be a witnessed model of $\mathcal{O}_\varphi$. Then there exists a function $g: \Delta^{\mathcal{I}_\varphi} \rightarrow \Delta^{\mathcal{I}}$ such that, for every $\mu \in \Delta^{\mathcal{I}_\varphi}$, $C^{\mathcal{I}_\varphi}(\mu) = C^{\mathcal{I}}(g(\mu))$ holds for every concept name C and $R_i^{\mathcal{I}_\varphi}(\mu, \mu i) = R_i^{\mathcal{I}}(g(\mu), g(\mu i))$ holds for every i , with $1 \leq i \leq p$.*

Proof. Let $\mathcal{I}$ be a witnessed model of $\mathcal{O}_\varphi$. We will build the function g inductively on the length of μ .

(ϵ) Since $\mathcal{I}$ is a model of $\mathcal{O}_\varphi$, then there is an element $\delta \in \Delta^{\mathcal{I}}$ such that $a^{\mathcal{I}} = \delta$. Since $\mathcal{I}$ is a model of $\mathcal{A}_\varphi$, setting $g(\epsilon) = \delta$, we have that $V^{\mathcal{I}_\varphi}(\epsilon) = 0 = V^{\mathcal{I}}(g(\epsilon))$ and the same holds for concept W . Moreover, since $\mathcal{I}$ is a model of $\mathcal{T}_\varphi$, we have that $V^{\mathcal{I}}(\delta) = (V_1 \sqcup V_2)^{\mathcal{I}}(\delta)$ and, therefore $V_1^{\mathcal{I}_\varphi}(\epsilon) = 0 = V_1^{\mathcal{I}}(g(\epsilon))$ and the same holds for V_2 , W_1 and W_2 . On the other hand, we have that $A^{\mathcal{I}_\varphi}(\epsilon) = 0.01 = A^{\mathcal{I}}(g(\epsilon))$, as well. Therefore, $g(\epsilon) = \delta$ satisfies the condition of the lemma.

(μi) Let now μ be such that $g(\mu)$ has already been defined. Now, since $\mathcal{I}$ is a witnessed model and satisfies axiom $\top \sqsubseteq \exists R_i \top$, then for all i , with $1 \leq i \leq p$, there exists a $\gamma \in \Delta^{\mathcal{I}}$ such that $R_i^{\mathcal{I}}(g(\mu), \gamma) = 1$. Setting $g(\mu i) = \gamma$ we get $1 = R_i^{\mathcal{I}_\varphi}(\mu, \mu i) = R_i^{\mathcal{I}}(g(\mu), g(\mu i))$. Furthermore, by inductive hypothesis, we can assume that $V^{\mathcal{I}}(g(\mu)) = 0.v_\mu$ and $W^{\mathcal{I}}(g(\mu)) = 0.w_\mu$.

Since $\mathcal{I}$ satisfies axiom $V \sqsubseteq (s+1)^{|v_i|} \cdot \forall R_i.V_1$, then

$$\begin{aligned} 0.v_\mu = V^{\mathcal{I}}(g(\mu)) &\leq (s+1)^{|v_i|} \cdot (\forall R_i.V_1)^{\mathcal{I}}(g(\mu)) = (s+1)^{|v_i|} \cdot \inf_{\gamma \in \Delta^{\mathcal{I}}} \{R_i^{\mathcal{I}}(g(\mu), \gamma) \Rightarrow V_1^{\mathcal{I}}(\gamma)\} \leq (s+1)^{|v_i|} \cdot R_i^{\mathcal{I}}(g(\mu), \mu i) \\ &\Rightarrow V_1^{\mathcal{I}}(\mu i) = (s+1)^{|v_i|} \cdot V_1^{\mathcal{I}}(g(\mu i)). \end{aligned}$$

Since $\mathcal{I}$ satisfies axiom $(s+1)^{|v_i|} \cdot \exists R_i.V_1 \sqsubseteq V$, then

$$\begin{aligned} 0.v_\mu = V^{\mathcal{I}}(g(\mu)) &\geq (s+1)^{|v_i|} \cdot (\exists R_i.V_1)^{\mathcal{I}}(g(\mu)) = (s+1)^{|v_i|} \cdot \sup_{\gamma \in \Delta^{\mathcal{I}}} \{R_i^{\mathcal{I}}(g(\mu), \gamma) \otimes V_1^{\mathcal{I}}(\gamma)\} \\ &\geq (s+1)^{|v_i|} \cdot R_i^{\mathcal{I}}(g(\mu), \mu i) \otimes V_1^{\mathcal{I}}(\mu i) = (s+1)^{|v_i|} \cdot V_1^{\mathcal{I}}(g(\mu i)). \end{aligned}$$

Therefore, $(s+1)^{|v_i|} \cdot V_1^{\mathcal{I}}(g(\mu i)) = 0.v_\mu$ and

$$V_1^{\mathcal{I}}(g(\mu i)) = 0.v_\mu \cdot (s+1)^{-|v_i|} = V_1^{\mathcal{I}_\varphi}(\mu i).$$

Similarly, it can be shown that $W_1^{\mathcal{I}}(g(\mu i)) = 0.w_\mu \cdot (s+1)^{-|w_i|} = W_1^{\mathcal{I}_\varphi}(\mu i)$.

Since $\mathcal{I}$ satisfies axioms $\langle \top \sqsubseteq \forall R_i.V_2, 0.v_i \rangle$ and $\langle \top \sqsubseteq \forall R_i.\neg V_2, 1 - 0.v_i \rangle$, it follows that $(\forall R_i.V_2)^{\mathcal{I}}(g(\mu)) \geq 0.v_i$ and $(\forall R_i.\neg V_2)^{\mathcal{I}}(g(\mu)) \geq 1 - 0.v_i$. Therefore, for $R_i^{\mathcal{I}}(g(\mu), g(\mu i)) = 1$ we have $V_2^{\mathcal{I}}(g(\mu i)) = 0.v_i = V_2^{\mathcal{I}_\varphi}(\mu i)$. Similarly, it can be shown that $W_2^{\mathcal{I}}(\mu i) = 0.w_i = W_2^{\mathcal{I}_\varphi}(\mu i)$.

Now, since $\mathcal{I}$ satisfies axiom $V \equiv V_1 \sqcup V_2$, then,

$$V^{\mathcal{I}}(g(\mu i)) = V_1^{\mathcal{I}}(g(\mu i)) + V_2^{\mathcal{I}}(g(\mu i)) = 0.v_\mu \cdot (s+1)^{-|v_i|} + 0.v_i = 0.v_i v_\mu = V^{\mathcal{I}_\varphi}(\mu i).$$

Finally, by inductive hypothesis, assume that

$$A^{\mathcal{I}}(g(\mu)) = A^{\mathcal{I}_\varphi}(\mu) = 0.01 \cdot (s+1)^{-\sum_{j \in \{i_1, i_2, \dots, i_k\}} \max\{|v_j|, |w_j|\|}},$$

where $\mu = i_1 i_2 \dots i_k$.

Since $\mathcal{I}$ satisfies axioms $A \sqsubseteq (s+1)^{\max\{|v_i|, |w_i|\|} \cdot \forall R_i.A$, we have that

$$A^{\mathcal{I}}(g(\mu)) \leq (s+1)^{\max\{|v_i|, |w_i|\|} \cdot (\forall R_i.A)^{\mathcal{I}}(g(\mu)) \leq (s+1)^{\max\{|v_i|, |w_i|\|} \cdot A^{\mathcal{I}}(g(\mu i)).$$

Likewise, since $\mathcal{I}$ satisfies axioms $(s+1)^{\max\{|v_i|, |w_i|\|} \cdot \exists R_i.A \sqsubseteq A$, we have that

$$A^{\mathcal{I}}(g(\mu)) \geq (s+1)^{\max\{|v_i|, |w_i|\|} \cdot (\exists R_i.A)^{\mathcal{I}}(g(\mu)) \geq (s+1)^{\max\{|v_i|, |w_i|\|} \cdot A^{\mathcal{I}}(g(\mu i))$$

and, thus,

$$A^{\mathcal{I}}(g(\mu)) = (s+1)^{\max\{|v_i|, |w_i|\|} \cdot A^{\mathcal{I}}(g(\mu i)).$$

Therefore,

$$\begin{aligned} A^{\mathcal{I}}(g(\mu i)) &= (s+1)^{-\max\{|v_i|, |w_i|\|} \cdot A^{\mathcal{I}}(g(\mu)) = (s+1)^{-\max\{|v_i|, |w_i|\|} \cdot A^{\mathcal{I}_\varphi}(\mu) \\ &= (s+1)^{-\max\{|v_i|, |w_i|\|} \cdot 0.01 \cdot (s+1)^{-\sum_{j \in \{i_1, i_2, \dots, i_k\}} \max\{|v_j|, |w_j|\|} = 0.01 \cdot (s+1)^{-\left(\max\{|v_i|, |w_i|\| + \sum_{j \in \{i_1, i_2, \dots, i_k\}} \max\{|v_j|, |w_j|\|}\right)} \\ &= 0.01 \cdot (s+1)^{-\sum_{j \in \{i_1, i_2, \dots, i_k, i\}} \max\{|v_j|, |w_j|\|} = A^{\mathcal{I}_\varphi}(\mu i), \end{aligned}$$

which completes the proof. $\square$

From the last Lemma it follows that if the RPCP instance φ has a solution μ , for some $\mu \in \{1, \dots, p\}^+$, then $v_\mu = w_\mu$ and, thus, $0.v_\mu = 0.w_\mu$. Therefore, every witnessed model $\mathcal{I}$ of $\mathcal{O}_\varphi$ contains an element $\delta = g(\mu)$ such that $V^\mathcal{I}(\delta) = V^{\mathcal{I}_\varphi}(\mu) = 0$. $v_\mu = 0.w_\mu = W^{\mathcal{I}_\varphi}(\mu) = W^\mathcal{I}(\delta)$. Conversely, from the definition of $\mathcal{I}_\varphi$, if φ has no solution, then there is no μ such that $0.v_\mu = 0.w_\mu$, i.e., there is no μ such that $V^{\mathcal{I}_\varphi}(\mu) = W^{\mathcal{I}_\varphi}(\mu)$.

However, as $\mathcal{O}_\varphi$ is always satisfiable, it does not yet help us to decide the RPCP. We next extend $\mathcal{O}_\varphi$ to $\mathcal{O}'_\varphi$ in such a way that an instance φ of the RPCP has a solution iff the ontology $\mathcal{O}'_\varphi$ is not witnessed satisfiable and, thus, establish that the KB satisfiability problem is undecidable. To this end, consider

$$\mathcal{O}'_\varphi := \langle \mathcal{T}'_\varphi, \mathcal{A} \rangle,$$

where

$$\mathcal{T}'_\varphi := \mathcal{T}_\varphi \cup \bigcup_{1 \leq i \leq p} \{ \top \sqsubseteq \forall R_i. (\neg(V \leftrightarrow W) \sqcup \neg A) \}.$$

The intuition here is the following. If there is a solution for RPCP then, by the observation before, there is a point δ in which the value of V and W coincide under $\mathcal{I}$. That is, the value of $\neg(V \leftrightarrow W)$ is 0 and, thus, the one of $\neg(V \leftrightarrow W) \sqcup \neg A$ is less than 1. Hence, $\mathcal{I}$ cannot satisfy the new GCI in $\mathcal{T}'_\varphi$ and, thus, $\mathcal{O}'_\varphi$ is not satisfiable. On the other hand, if there is no solution to the RPCP then in $\mathcal{I}_\varphi$ there is no point in which V and W coincide and, thus, $\neg(V \leftrightarrow W) > 0$. Moreover, we will show that the value of $\neg(V \leftrightarrow W)$ in all points is strictly greater than A and, as $A \sqcup \neg A$ is 1, so also $\neg(V \leftrightarrow W) \sqcup \neg A$ will be 1 in any point. Hence, $\mathcal{I}_\varphi$ is a model of the additional axiom in $\mathcal{T}'_\varphi$, i.e., $\mathcal{O}'_\varphi$ is satisfiable.

Proposition 6. *The instance φ of the RPCP has a solution iff the ontology $\mathcal{O}'_\varphi$ is not witnessed satisfiable.*

Proof. Assume first that φ has a solution $\mu = i_1 \dots i_k$ and let $\mathcal{I}$ be a witnessed model of $\mathcal{O}_\varphi$. Let $\bar{\mu} = i_1 i_2 \dots i_{k-1}$ (last index i_k is dropped from μ). Then by Lemma 5 it follows that there are nodes $\delta, \delta' \in \Delta^\mathcal{I}$ such that $\delta = g(\mu)$, $\delta' = g(\bar{\mu})$, with $V^\mathcal{I}(\delta) = V^{\mathcal{I}_\varphi}(\mu) = W^{\mathcal{I}_\varphi}(\mu) = W^\mathcal{I}(\delta)$ and $R_{i_k}^\mathcal{I}(\delta', \delta) = 1$. Then $(V \leftrightarrow W)^\mathcal{I}(\delta) = 1$. Since $(\neg A)^\mathcal{I}(\delta) < 1$, then $(\neg(V \leftrightarrow W) \sqcup \neg A)^\mathcal{I}(\delta) < 1$. Hence there is i , with $1 \leq i \leq p$, such that $(\forall R_i. (\neg(V \leftrightarrow W) \sqcup \neg A))^\mathcal{I}(\delta') < 1$. Therefore, axiom $\top \sqsubseteq \forall R_i. (\neg(V \leftrightarrow W) \sqcup \neg A)$ is not satisfied and, then, $\mathcal{O}'_\varphi$ is not satisfiable.

For the converse, assume that φ has no solution. On the one hand we know that $\mathcal{I}_\varphi$ is a model of $\mathcal{O}_\varphi$. On the other hand, since φ has no solution, then there is no $\mu = i_1 \dots i_k$ such that $v_\mu = w_\mu$ (i.e., $0.v_\mu = 0.w_\mu$) and, therefore, there is no $\mu \in \Delta^{\mathcal{I}_\varphi}$ such that $V^{\mathcal{I}_\varphi}(\mu) = W^{\mathcal{I}_\varphi}(\mu)$. Consider $\mu \in \Delta^{\mathcal{I}_\varphi}$ and i , with $1 \leq i \leq p$ and assume, without loss of generality, that $V^{\mathcal{I}_\varphi}(\mu i) < W^{\mathcal{I}_\varphi}(\mu i)$. Then

$$\begin{aligned} (V \leftrightarrow W)^{\mathcal{I}_\varphi}(\mu i) &= (V^{\mathcal{I}_\varphi}(\mu i) \Rightarrow W^{\mathcal{I}_\varphi}(\mu i)) \otimes (W^{\mathcal{I}_\varphi}(\mu i) \Rightarrow V^{\mathcal{I}_\varphi}(\mu i)) = 1 \otimes (W^{\mathcal{I}_\varphi}(\mu i) \Rightarrow V^{\mathcal{I}_\varphi}(\mu i)) = W^{\mathcal{I}_\varphi}(\mu i) \Rightarrow V^{\mathcal{I}_\varphi}(\mu i) \\ &= 1 - W^{\mathcal{I}_\varphi}(\mu i) + V^{\mathcal{I}_\varphi}(\mu i) = 1 - (W^{\mathcal{I}_\varphi}(\mu i) - V^{\mathcal{I}_\varphi}(\mu i)) = 1 - (0.w_{\mu i} - 0.v_{\mu i}) \leq 1 - 0.01 \cdot (s+1)^{-\max\{|v_{\mu i}|, |w_{\mu i}|\}} \\ &\quad - \sum_{j \in \{i_1, i_2, \dots, i_k, i\}} \max\{|v_j|, |w_j|\} \\ &\leq 1 - 0.01 \cdot (s+1) = (\neg A)^{\mathcal{I}_\varphi}(\mu i). \end{aligned}$$

Therefore, $(\neg(V \leftrightarrow W))^{\mathcal{I}_\varphi}(\mu i) \geq A^{\mathcal{I}_\varphi}(\mu i)$. As $A^{\mathcal{I}_\varphi}(\mu i) \oplus (\neg A)^{\mathcal{I}_\varphi}(\mu i) = 1$, it follows that for every $\mu \in \Delta^{\mathcal{I}_\varphi}$ and i , with $1 \leq i \leq p$, it holds that $(\forall R_i. (\neg(V \leftrightarrow W) \sqcup \neg A))^{\mathcal{I}_\varphi}(\mu) = 1$ and, therefore, $\mathcal{I}_\varphi$ is a witnessed model of $\mathcal{O}'_\varphi$. $\square$

By Proposition 6, we have a reduction of a RPCP to a KB satisfiability problem. Note that all roles are crisp. Therefore,

Proposition 7. *The knowledge base satisfiability problem is undecidable for $\mathbb{L}$ -ALC with GCIs. The result holds also if crisp roles are assumed.*

Undecidability of KB satisfiability w.r.t. finite models. In this section we address a sub problem of the previous one. That is, deciding whether a KB has a finite model.

As in [7], given an instance φ of RPCP, we provide an ontology $\tilde{\mathcal{O}}_\varphi$ and prove that it has a finite model iff φ has a solution. We now define a TBox $\tilde{\mathcal{T}}$ as follows:

$$\tilde{\mathcal{T}} := \{ V \equiv V_1 \sqcup V_2, W \equiv W_1 \sqcup W_2, \neg(V \leftrightarrow W) \sqsubseteq \max\{C_1, \dots, C_p\} \},$$

and TBoxes $\tilde{\mathcal{T}}_\varphi^i$ as follows:

$$\begin{aligned} \tilde{\mathcal{T}}_\varphi^i &:= \{ C_i \equiv \exists R_i. \top, \top \sqsubseteq \max\{C_i, \neg C_i\}, (C_i \rightarrow V) \sqsubseteq (s+1)^{|v_i|} \cdot \forall R_i. V_1, (s+1)^{|v_i|} \cdot \exists R_i. V_1 \sqsubseteq (C_i \rightarrow V), (C_i \rightarrow W) \\ &\quad \sqsubseteq (s+1)^{|w_i|} \cdot \forall R_i. W_1, (s+1)^{|w_i|} \cdot \exists R_i. W_1 \sqsubseteq (C_i \rightarrow W), \langle \top \sqsubseteq \forall R_i. V_2, 0.v_i \rangle, \langle \top \sqsubseteq \forall R_i. \neg V_2, 1 - 0.v_i \rangle, \\ &\quad \langle \top \sqsubseteq \forall R_i. W_2, 0.w_i \rangle, \langle \top \sqsubseteq \forall R_i. \neg W_2, 1 - 0.w_i \rangle \} \end{aligned}$$

Now, let

$$\tilde{\mathcal{T}}_\varphi = \tilde{\mathcal{T}} \cup \bigcup_{i=1}^p \tilde{\mathcal{T}}_\varphi^i.$$

Further we define the ABox $\tilde{\mathcal{A}}_\varphi$ as follows:

$$\tilde{\mathcal{A}}_\varphi := a : \neg V, a : \neg W, a : \max\{C_1, \dots, C_p\}.$$

Finally,

$$\tilde{\mathcal{O}}_\varphi := \langle \tilde{\mathcal{T}}_\varphi, \tilde{\mathcal{A}}_\varphi \rangle.$$

Proposition 8. *The instance φ of the RPCP has a solution iff the ontology $\tilde{\mathcal{O}}_\varphi$ has a finite model.*

Proof.

($\Rightarrow$) Let $\mu = i_1 \dots i_k$ be a solution of φ and let $\text{suff}(\mu)$ be the set of all suffixes of μ .⁴ We build the finite interpretation $\tilde{\mathcal{I}}_\varphi$ as follows:

- $\Delta^{\tilde{\mathcal{I}}_\varphi} := \text{suff}(\mu)$,
- $\mathbf{a}^{\tilde{\mathcal{I}}_\varphi} = \epsilon$,
- $V^{\tilde{\mathcal{I}}_\varphi}(\epsilon) = W^{\tilde{\mathcal{I}}_\varphi}(\epsilon) = 0$, and for $1 \leq i \leq 2$, $V_i^{\tilde{\mathcal{I}}_\varphi}(\epsilon) = W_i^{\tilde{\mathcal{I}}_\varphi}(\epsilon) = 0$,
- for all $v \in \Delta^{\tilde{\mathcal{I}}_\varphi}$, $V^{\tilde{\mathcal{I}}_\varphi}(v) = 0.v_v$, $W^{\tilde{\mathcal{I}}_\varphi}(v) = 0.w_v$,
- for all $v, v' \in \Delta^{\tilde{\mathcal{I}}_\varphi}$ and $1 \leq i \leq p$,

$$R_i^{\tilde{\mathcal{I}}_\varphi}(v, v') = \begin{cases} 1, & \text{if } v' = iv, \\ 0, & \text{otherwise,} \end{cases}$$

- for all $v \in \Delta^{\tilde{\mathcal{I}}_\varphi}$ and $1 \leq i \leq p$,

$$C_i^{\tilde{\mathcal{I}}_\varphi}(v) = \begin{cases} 1, & \text{if } iv \in \text{suff}(\mu), \\ 0, & \text{otherwise,} \end{cases}$$

- for all $v \in \Delta^{\tilde{\mathcal{I}}_\varphi}$ and $1 \leq i \leq p$ such that $iv \in \text{suff}(\mu)$
 - $V_1^{\tilde{\mathcal{I}}_\varphi}(iv) = 0.v_v \cdot (s+1)^{-|v_i|}$, $W_1^{\tilde{\mathcal{I}}_\varphi}(iv) = 0.w_v \cdot (s+1)^{-|w_i|}$,
 - $V_2^{\tilde{\mathcal{I}}_\varphi}(iv) = 0.v_i$, $W_2^{\tilde{\mathcal{I}}_\varphi}(iv) = 0.w_i$.

We show now that $\tilde{\mathcal{I}}_\varphi$ is a model $\tilde{\mathcal{O}}_\varphi$. Since $V^{\tilde{\mathcal{I}}_\varphi}(\epsilon) = 0.v_\epsilon = 0$ and $W^{\tilde{\mathcal{I}}_\varphi}(\epsilon) = 0.w_\epsilon = 0$, then the first two axioms in $\tilde{\mathcal{A}}_\varphi$ are satisfied. Since there is $1 \leq i \leq p$ such that $i\epsilon = i \in \text{suff}(\mu)$, then $C_i^{\tilde{\mathcal{I}}_\varphi}(\epsilon) = 1$ and, therefore, the third axiom in $\tilde{\mathcal{A}}_\varphi$ is satisfied.

We now show that the axioms in $\tilde{\mathcal{T}}$ and each $\tilde{\mathcal{T}}_\varphi^i$, with $1 \leq i \leq p$ are satisfied for every $v \in \text{suff}(\mu)$. Let $v \in \text{suff}(\mu) \setminus \{\mu\}$. Then there is

$1 \leq i \leq p$ such that $iv \in \text{suff}(\mu)$ and, therefore, by the definition of $\tilde{\mathcal{I}}_\varphi$, $C_i^{\tilde{\mathcal{I}}_\varphi}(v) = 1$ and $R_i^{\tilde{\mathcal{I}}_\varphi}(v, iv) = 1$. Therefore, $(C_i \rightarrow V)^{\tilde{\mathcal{I}}_\varphi}(v) = V^{\tilde{\mathcal{I}}_\varphi}(v)$ from which it follows that every axiom in $\tilde{\mathcal{T}}_\varphi^i$ is satisfied by $\tilde{\mathcal{I}}_\varphi$ (the proof is the same as for $\mathcal{I}_\varphi$ satisfying $\mathcal{T}_\varphi^i$). E.g.,

note that $V^{\tilde{\mathcal{I}}_\varphi}(v) = 0.v_v = (s+1)^{|v_i|} \cdot V_1^{\tilde{\mathcal{I}}_\varphi}(iv)$ and, thus, both $(C_i \rightarrow V) \sqsubseteq (s+1)^{|v_i|} \cdot \forall R_i.V_1$ and $(s+1)^{|v_i|} \cdot \exists R_i.V_1 \sqsubseteq (C_i \rightarrow V)$ are satisfied.

Moreover, for every $j \neq i$ and $v' \in \text{suff}(\mu)$, it holds that $C_j^{\tilde{\mathcal{I}}_\varphi}(v) = 0$ and $R_j^{\tilde{\mathcal{I}}_\varphi}(v, v') = 0$ and, therefore every axiom in $\tilde{\mathcal{T}}_\varphi^j$ is satisfied as well (note that e.g., $(\forall R_j.V_1)^{\tilde{\mathcal{I}}_\varphi}(v) = 1$). This last argument holds for μ as well.

Finally, consider $\tilde{\mathcal{T}}_\varphi$. It is easy to check that the first two axioms are satisfied in every $v \in \text{suff}(\mu)$. For the third axiom, if $v \in \text{suff}(\mu) \setminus \{\mu\}$, then there is $1 \leq i \leq p$ such that $C_i^{\tilde{\mathcal{I}}_\varphi}(v) = 1$ and, then, the axiom is trivially satisfied. Otherwise, if $v = \mu$, since μ is a solution for φ , then $(\neg(V \leftrightarrow W))^{\tilde{\mathcal{I}}_\varphi}(\mu) = 0$ and, then, the axiom is trivially satisfied as well.

($\Leftarrow$) For the converse, suppose that φ has no solution and let $\mathcal{I}$ be a model of $\tilde{\mathcal{O}}_\varphi$. By absurd, let us assume that $\mathcal{I}$ is finite and, thus, witnessed.

Now, since $\mathcal{I}$ is a model of axioms $a : \neg V$ and $a : \neg W$, then there is a node $a^\mathcal{I} = \delta \in \Delta^\mathcal{I}$, such that $V^\mathcal{I}(\delta) = W^\mathcal{I}(\delta) = 0$.

Moreover, since $\mathcal{I}$ is a model of axioms $V \equiv V_1 \sqcup V_2$ and $W \equiv W_1 \sqcup W_2$, then $V_1^\mathcal{I}(\delta) = V_2^\mathcal{I}(\delta) = W_1^\mathcal{I}(\delta) = W_2^\mathcal{I}(\delta) = 0$ as well.

Next, we prove by induction that for every $n \in \mathbb{N}$ there is an element $\delta_n \in \Delta^\mathcal{I}$ such that:

⁴ A suffix of a string $t_1 t_2 \dots t_n$ is a string $t_{n-m+1} \dots t_n$ ($0 \leq m \leq n$), which is the empty string ϵ for $m = 0$.

- $V^{\mathcal{I}}(\delta_{i_n}) = 0.v_{i_n} \dots v_{i_1}$,
- $W^{\mathcal{I}}(\delta_{i_n}) = 0.w_{i_n} \dots w_{i_1}$,

and $|\{\delta, \delta_{i_1}, \dots, \delta_{i_n}\}| = n+1$ (all elements are distinct). As a consequence, $\mathcal{A}^{\mathcal{I}}$ cannot be finite, contrary to the assumption that $\mathcal{I}$ is finite.

Case $n = 1$. Since $\mathcal{I}$ is a witnessed model, it satisfies axiom $a: \max\{C_1, \dots, C_p\}$. Hence, there is i , such that $C_i^{\mathcal{I}}(\delta) = 1$. Let $i_1 = i$. Since $\mathcal{I}$ satisfies axiom $C_{i_1} \equiv \exists R_{i_1}. \top$, then there is $\delta' \in \mathcal{A}^{\mathcal{I}}$ such that $R_{i_1}^{\mathcal{I}}(\delta, \delta') = 1$. Let $\delta_{i_1} = \delta'$. Since $\mathcal{I}$ satisfies axiom $(s+1)^{|v_{i_1}|} \cdot \exists R_{i_1}. V_1 \sqsubseteq (C_{i_1} \rightarrow V)$, then

$$0 = (1 \Rightarrow 0) = (C_{i_1}(\delta) \Rightarrow V)^{\mathcal{I}}(\delta) \geq (s+1)^{|v_{i_1}|} \cdot \sup_{\delta' \in \mathcal{A}^{\mathcal{I}}} \{R_{i_1}^{\mathcal{I}}(\delta, \delta') \otimes V_1^{\mathcal{I}}(\delta')\} \geq R_{i_1}^{\mathcal{I}}(\delta, \delta_{i_1}) \otimes V_1^{\mathcal{I}}(\delta_{i_1}) = 1 \otimes V_1^{\mathcal{I}}(\delta_{i_1}) = V_1^{\mathcal{I}}(\delta_{i_1}).$$

Hence, $V_1^{\mathcal{I}}(\delta_{i_1}) = 0$. In the same way it can be proven that $W_1^{\mathcal{I}}(\delta_{i_1}) = 0$.

Since $\mathcal{I}$ satisfies axiom $\langle \top \sqsubseteq \forall R_{i_1}. V_2, 0.v_{i_1} \rangle$, we have that $0.v_{i_1} \leq (R_{i_1}^{\mathcal{I}}(\delta, \delta_{i_1}) \Rightarrow V_2^{\mathcal{I}}(\delta_{i_1})) = (1 \Rightarrow V_2^{\mathcal{I}}(\delta_{i_1})) = V_2^{\mathcal{I}}(\delta_{i_1})$.

Since $\mathcal{I}$ satisfies axiom $\langle \top \sqsubseteq \forall R_{i_1}. \neg V_2, 1 - 0.v_{i_1} \rangle$, it follows that $1 - 0.v_{i_1} \leq (R_{i_1}^{\mathcal{I}}(\delta, \delta_{i_1}) \Rightarrow \neg V_2^{\mathcal{I}}(\delta_{i_1})) = (1 \Rightarrow \neg V_2^{\mathcal{I}}(\delta_{i_1})) = \neg V_2^{\mathcal{I}}(\delta_{i_1}) = 1 - V_2^{\mathcal{I}}(\delta_{i_1})$ and therefore, $V_2^{\mathcal{I}}(\delta_{i_1}) \leq 0.v_{i_1}$. Therefore, $V_2^{\mathcal{I}}(\delta_{i_1}) = 0.v_{i_1}$. In the same way it can be proven that $W_2^{\mathcal{I}}(\delta_{i_1}) = 0.w_{i_1}$.

Finally, since $\mathcal{I}$ satisfies axiom $V \equiv V_1 \sqcup V_2$, then $V^{\mathcal{I}}(\delta_{i_1}) = V_1^{\mathcal{I}}(\delta_{i_1}) \oplus V_2^{\mathcal{I}}(\delta_{i_1}) = 0 \oplus 0.v_{i_1} = 0.v_{i_1}$. In the same way it can be proven that $W^{\mathcal{I}}(\delta_{i_1}) = 0.w_{i_1}$. Moreover, since $V^{\mathcal{I}}(\delta) = 0 \neq 0.v_{i_1} = V^{\mathcal{I}}(\delta_{i_1})$, then $\delta \neq \delta_{i_1}$ and, thus, $|\{\delta, \delta_{i_1}\}| = 2$, which completes the case.

Induction step $n+1$. Let $n > 1$ and suppose, by inductive hypothesis, that, for every $j \leq n$, the above conditions hold.

Since φ has no solution, then $v_{i_n} \dots v_{i_1} \neq w_{i_n} \dots w_{i_1}$ and, therefore, by inductive hypothesis, $V^{\mathcal{I}}(\delta_{i_n}) = 0$. $v_{i_n} \dots v_{i_1} \neq 0.w_{i_n} \dots w_{i_1} = W^{\mathcal{I}}(\delta_{i_n})$. Hence $(V \leftrightarrow W)^{\mathcal{I}}(\delta_{i_n}) < 1$ and, therefore, $\neg(V \leftrightarrow W)^{\mathcal{I}}(\delta_{i_n}) > 0$. Now, since $\mathcal{I}$ satisfies axiom $\neg(V \leftrightarrow W) \sqsubseteq \max\{C_1, \dots, C_p\}$, $(\max\{C_1, \dots, C_p\})^{\mathcal{I}}(\delta_{i_n}) > 0$ follows and, thus, there is i such that $C_i^{\mathcal{I}}(\delta_{i_n}) > 0$. Therefore, as $\mathcal{I}$ satisfies axiom $\top \sqsubseteq \max\{C_i, \neg C_i\}$, we have that $C_i^{\mathcal{I}}(\delta_{i_n}) = 1$. Now, let $i_{n+1} = i$.

Since $\mathcal{I}$ satisfies axiom $C_{i_{n+1}} \equiv \exists R_{i_{n+1}}. \top$, then there is $\delta' \in \mathcal{A}^{\mathcal{I}}$ such that $R_{i_{n+1}}^{\mathcal{I}}(\delta_{i_n}, \delta') = 1$. Let $\delta_{i_{n+1}} = \delta'$.

Since $\mathcal{I}$ satisfies axiom $(C_{i_{n+1}} \rightarrow V) \sqsubseteq (s+1)^{|v_{i_{n+1}}|} \cdot \forall R_{i_{n+1}}. V_1$, then

$$\begin{aligned} 0.v_{i_n} \dots v_{i_1} &= (1 \Rightarrow 0.v_{i_n} \dots v_{i_1}) = (C_{i_n} \Rightarrow V)^{\mathcal{I}}(\delta_{i_n}) \leq (s+1)^{|v_{i_{n+1}}|} \cdot \inf_{\delta' \in \mathcal{A}^{\mathcal{I}}} \{R_{i_{n+1}}^{\mathcal{I}}(\delta_{i_n}, \delta') \Rightarrow V_1^{\mathcal{I}}(\delta')\} \\ &\leq (R_{i_{n+1}}^{\mathcal{I}}(\delta_{i_n}, \delta_{i_{n+1}}) \Rightarrow V_1^{\mathcal{I}}(\delta_{i_{n+1}})) = V_1^{\mathcal{I}}(\delta_{i_{n+1}}). \end{aligned}$$

On the other hand, since $\mathcal{I}$ satisfies axiom $(s+1)^{|v_{i_{n+1}}|} \cdot \exists R_{i_{n+1}}. V_1 \sqsubseteq (C_{i_{n+1}} \rightarrow V)$, then

$$\begin{aligned} 0.v_{i_n} \dots v_{i_1} &= (1 \Rightarrow 0.v_{i_n} \dots v_{i_1}) = (C_{i_n} \Rightarrow V)^{\mathcal{I}}(\delta_{i_n}) \geq (s+1)^{|v_{i_{n+1}}|} \cdot \sup_{\delta' \in \mathcal{A}^{\mathcal{I}}} \{R_{i_{n+1}}^{\mathcal{I}}(\delta_{i_n}, \delta') \otimes V_1^{\mathcal{I}}(\delta')\} \\ &\geq (s+1)^{|v_{i_{n+1}}|} \cdot (R_{i_{n+1}}^{\mathcal{I}}(\delta_{i_n}, \delta_{i_{n+1}}) \otimes V_1^{\mathcal{I}}(\delta_{i_{n+1}})) = (s+1)^{|v_{i_{n+1}}|} \cdot V_1^{\mathcal{I}}(\delta_{i_{n+1}}). \end{aligned}$$

Hence, $0.v_{i_n} \dots v_{i_1} = (s+1)^{|v_{i_{n+1}}|} \cdot V_1^{\mathcal{I}}(\delta_{i_{n+1}})$ and, thus,

$$V_1^{\mathcal{I}}(\delta_{i_{n+1}}) = (s+1)^{-|v_{i_{n+1}}|} \cdot 0.v_{i_n} \dots v_{i_1}.$$

In the same way it can be proven that $W_1^{\mathcal{I}}(\delta_{i_{n+1}}) = 0.w_{i_n} \dots w_{i_1} \cdot (s+1)^{-|w_{i_{n+1}}|}$. Since $\mathcal{I}$ satisfies axiom $\langle \top \sqsubseteq \forall R_{i_{n+1}}. V_2, 0.v_{i_{n+1}} \rangle$, we get

$$0.v_{i_{n+1}} \leq R_{i_{n+1}}^{\mathcal{I}}(\delta_{i_n}, \delta_{i_{n+1}}) \Rightarrow V_2^{\mathcal{I}}(\delta_{i_{n+1}}) = 1 \Rightarrow V_2^{\mathcal{I}}(\delta_{i_{n+1}}) = V_2^{\mathcal{I}}(\delta_{i_{n+1}}).$$

Similarly, since $\mathcal{I}$ satisfies axiom $\langle \top \sqsubseteq \forall R_{i_{n+1}}. \neg V_2, 1 - 0.v_{i_{n+1}} \rangle$, we get

$$1 - 0.v_{i_{n+1}} \leq R_{i_{n+1}}^{\mathcal{I}}(\delta_{i_n}, \delta_{i_{n+1}}) \Rightarrow \neg V_2^{\mathcal{I}}(\delta_{i_{n+1}}) = 1 \Rightarrow \neg V_2^{\mathcal{I}}(\delta_{i_{n+1}}) = \neg V_2^{\mathcal{I}}(\delta_{i_{n+1}}) = 1 - V_2^{\mathcal{I}}(\delta_{i_{n+1}})$$

and therefore, $V_2^{\mathcal{I}}(\delta_{i_{n+1}}) \leq 0.v_{i_{n+1}}$. Hence,

$$V_2^{\mathcal{I}}(\delta_{i_{n+1}}) = 0.v_{i_{n+1}}.$$

In the same way it can be proven that $W_2^{\mathcal{I}}(\delta_{i_{n+1}}) = 0.w_{i_{n+1}}$.

Finally, since $\mathcal{I}$ satisfies axiom $V \equiv V_1 \sqcup V_2$, then

$$V^{\mathcal{I}}(\delta_{i_{n+1}}) = V_1^{\mathcal{I}}(\delta_{i_{n+1}}) \oplus V_2^{\mathcal{I}}(\delta_{i_{n+1}}) = ((s+1)^{-|v_{i_{n+1}}|} \cdot 0.v_{i_n} \dots v_{i_1}) \oplus 0.v_{i_{n+1}} = 0.v_{i_{n+1}} \dots 0.v_{i_1}.$$

In the same way it can be proven that $W^{\mathcal{I}}(\delta_{i_{n+1}}) = 0.w_{i_{n+1}} \dots 0.w_{i_1}$.

Moreover, since, by inductive hypothesis, for every $j \leq n$,

$$V^{\mathcal{I}}(\delta_{i_j}) = 0.v_{i_j} \dots v_{i_1} \neq 0.v_{i_{n+1}} \dots v_{i_j} \dots v_{i_1} = V^{\mathcal{I}}(\delta_{i_{n+1}}),$$

then $\delta_{i_j} \neq \delta_{i_{n+1}}$. Furthermore, as $V^T(\delta) = 0 \neq V^T(\delta_{i_{n+1}})$, then $\delta \neq \delta_{i_{n+1}}$ and, thus, $|\{\delta, \delta_{i_1}, \dots, \delta_{i_{n+1}}\}| = n + 2$, which completes the case.

Hence, $\tilde{\mathcal{O}}_\varphi$ has no finite model. $\square$

By Proposition 8, we have a reduction of a RPCP to a finite satisfiability problem. Again, note that all roles are crisp. Therefore,

Proposition 9. *The knowledge base finite satisfiability problem is undecidable for $\mathbb{L}$ -ALC with GCIs. The result holds also if crisp roles are assumed.*

We conclude by pointing out that, as $\mathcal{K} \models \langle a : \perp, 1 \rangle$ iff $\mathcal{K}$ is not satisfiable iff $\mathcal{K} \models \langle \top \sqsubseteq \perp, 1 \rangle$, both the entailment problem of determining whether $\mathcal{K} \models \langle a : C, n \rangle$ and the problem of determining whether $\mathcal{K} \models \langle C \sqsubseteq D, n \rangle$ are undecidable, and, thus, as well are undecidable the problems of determining $bed(\mathcal{K}, a : C)$ and $bsd(\mathcal{K}, C)$ (w.r.t. arbitrary witnessed or finite models).

Corollary 10. *For $\mathbb{L}$ -ALC with GCIs, with respect to arbitrary witnessed or finite models, (i) the best entailment degree problem for concept assertions and GCIs is undecidable; and (ii) the best satisfiability degree problem is undecidable. These results hold also if crisp roles are assumed.*

4. A MILP based decision procedure for ALC under Łukasiewicz logic

In the following, given a knowledge base $\mathcal{K} = \langle \mathcal{T}, \mathcal{A} \rangle$, w.l.o.g. we will assume that GCIs in $\mathcal{T}$ are of the form $\langle \top \sqsubseteq C, n \rangle$ and that all concepts in $\mathcal{K}$ are in *Negation Normal Form* (NNF), i.e., only concept names can be negated. This can be easily achieved by using the following transformation rules

$$\begin{aligned} \neg\neg C &\mapsto C, \\ \neg(C \sqcap D) &\mapsto \neg C \sqcup \neg D, \\ \neg(C \sqcup D) &\mapsto \neg C \sqcap \neg D, \\ \neg\exists R.C &\mapsto \forall R.\neg C, \\ \neg\forall R.C &\mapsto \exists R.\neg C. \end{aligned}$$

Let us also introduce a restricted form of TBoxes, i.e., *acyclic* TBoxes. Specifically, let us consider GCIs of one of the following forms

$$\begin{aligned} \langle \top \sqsubseteq A \rightarrow C, n \rangle, \\ \langle \top \sqsubseteq A \leftrightarrow C, n \rangle, \end{aligned}$$

where A is a concept name and C is a concept. We call a GCI of the form $\langle \top \sqsubseteq A \rightarrow C, n \rangle$ *primitive* and call $\langle \top \sqsubseteq A \leftrightarrow C, n \rangle$ *definitional*. We say that A is the head of these axioms and C is the body.

Now, we say that

- concept name A *directly uses* concept name B w.r.t. $\mathcal{T}$, denoted $A \rightarrow_{\mathcal{T}} B$, if A is the head of some axiom $\tau \in \mathcal{T}$ such that B occurs in the body of τ ;
- concept name A *uses* concept name B w.r.t. $\mathcal{T}$, denoted $A \rightsquigarrow_{\mathcal{T}} B$, if there exist concept names $A_1, \dots, A_n$, such that $A_1 = A$, $A_n = B$ and, for every $1 \leq i < n$, it holds that $A_i \rightarrow_{\mathcal{T}} A_{i+1}$.

Eventually, we say that a TBox $\mathcal{T}$ is *acyclic* [3] iff

1. every GCI is either primitive or definitional,
2. there is no A such that $A \rightsquigarrow_{\mathcal{T}} A$.
3. no concept name A is in the head of more than one axiom.

Note that restriction 3. allows to avoid that a GCI $\langle C \sqsubseteq D, n \rangle$ may be introduced indirectly, e.g., via

$$\begin{aligned} \langle \top \sqsubseteq A \leftrightarrow C, 1 \rangle, \\ \langle \top \sqsubseteq B \leftrightarrow D, 1 \rangle, \\ \langle \top \sqsubseteq A \rightarrow B, n \rangle. \end{aligned}$$

Now, we recall from [8] the following property.

Proposition 11 [8] Theorem 4.4. *An acyclic $\mathbb{L}$ -ALC knowledge base $\mathcal{K}$ is satisfiable iff it has a finite model.*

Therefore, by Proposition 11 we may restrict our attention to finite models only, which indeed will be the case in the following Sections 4.1 and 4.3.

We first describe a sound, complete and terminating MILP based decision procedure for $\mathcal{L}\text{-}\mathcal{ALC}$ without TBoxes, i.e., only with ABox. Subsequently we extend the procedure in order to cope with acyclic TBoxes as well.

The procedure is a slightly variant of and conceptually identical to the ones described in previous works such as [12–15,52].

Our proofs showing that the ABox and acyclic KB satisfiability problems are decidable consist both of the following steps:

1. first, we introduce the notion of fuzzy tableau, which is a mathematical structure with the property that a knowledge base is satisfiable iff there is a fuzzy tableau; since the notion of fuzzy tableau is the same for both problems, we define it in a framework that is common in Section 4.1.
2. second, we provide a terminating algorithm that, starting from an ABox (acyclic KB, respectively), builds a forest of trees (called completion forest) together with a MILP problem. We will show that the generated MILP problem has a solution iff there is a fuzzy tableau for the ABox (acyclic KB, respectively) at hand, which together with Point 1. above completes the proof.

4.1. Fuzzy tableaux

Let us start with the definition of fuzzy tableau.

Definition 12. A *fuzzy tableau* T for $\mathcal{K} = \langle \mathcal{T}, \mathcal{A} \rangle$ is a quadruple $(\mathbf{S}, \mathcal{L}, \mathcal{E}, \mathcal{V})$ such that: $\mathbf{S}$ is a set of elements, $\mathcal{L} : \mathbf{S} \times \text{sub}(\mathcal{K}) \rightarrow [0, 1]$ maps each element and concept, to a membership degree (the degree of the element being an instance of the concept), and $\mathcal{E} : \mathbf{R}_{\mathcal{K}} \times (\mathbf{S} \times \mathbf{S}) \rightarrow [0, 1]$ maps each role of $\mathbf{R}_{\mathcal{K}}$ (the set of roles occurring in $\mathcal{K}$) and pair of elements to the membership degree of the pair being an instance of the role, and $\mathcal{V} : \mathbf{I}_{\mathcal{A}} \rightarrow \mathbf{S}$ maps individuals occurring in $\mathcal{A}$ to elements in $\mathbf{S}$ (where $\mathcal{V}(a) \neq \mathcal{V}(b)$ if $a \neq b$). A fuzzy tableau has to satisfy the following properties reflecting the semantics of the connectives in Łukasiewicz logic. For all $s \in \mathbf{S}$, $C_1, C_2 \in \text{sub}(\mathcal{K})$ and $R \in \mathbf{R}_{\mathcal{K}}$, T has to satisfy:

1. $\mathcal{L}(s, \perp) = 0$ and $\mathcal{L}(s, \top) = 1$ for all $s \in \mathbf{S}$;
2. $\mathcal{L}(s, \neg A) \leq 1 - \mathcal{L}(s, A)$;
3. $\mathcal{L}(s, C_1 \sqcap C_2) \leq \mathcal{L}(s, C_1) \otimes \mathcal{L}(s, C_2)$;
4. $\mathcal{L}(s, C_1 \sqcup C_2) \leq \mathcal{L}(s, C_1) \oplus \mathcal{L}(s, C_2)$;
5. for all $t \in \mathbf{S}$, $\mathcal{L}(t, C) \geq \mathcal{E}(R, \langle s, t \rangle) \otimes \mathcal{L}(s, \forall R.C)$;
6. there is $t \in \mathbf{S}$ such that $\mathcal{L}(s, \exists R.C) \leq \mathcal{E}(R, \langle s, t \rangle) \otimes \mathcal{L}(t, C)$;
7. if $\langle a : C, n \rangle \in \mathcal{A}$, then $\mathcal{L}(\mathcal{V}(a), C) \geq n$;
8. if $\langle \langle a, b \rangle : R, n \rangle \in \mathcal{A}$, then $\mathcal{E}(R, \langle \mathcal{V}(a), \mathcal{V}(b) \rangle) \geq n$;
9. if $\langle \top \sqsubseteq C, n \rangle \in \mathcal{T}$, then for all $s \in \mathbf{S}$, $\mathcal{L}(s, C) \geq n$.

The conditions above are based on the semantics of the connectives $\sqcap$, $\sqcup$, $\forall$ and $\exists$. Informally, our aim is to build from a fuzzy tableau an interpretation $\mathcal{I}$ such that whenever $\mathcal{L}(s, C) \geq n$ then $C^{\mathcal{I}}(s) \geq n$, and vice versa.

The following Proposition can be proven.

Proposition 13. A fuzzy knowledge base $\mathcal{K}$ has a finite model iff there is a finite fuzzy tableau for $\mathcal{K}$.

Proof. Let $\mathcal{K}$ be a fuzzy knowledge base, then:

($\Leftarrow$) For the *if* direction let $T = (\mathbf{S}, \mathcal{L}, \mathcal{E}, \mathcal{V})$ be a finite fuzzy tableau for $\mathcal{K}$, then we can construct a fuzzy interpretation $\mathcal{I} = (\mathcal{A}^{\mathcal{I}}, \cdot^{\mathcal{I}})$ as follows:

$$\begin{aligned} \mathcal{A}^{\mathcal{I}} &= \mathbf{S}, \quad a^{\mathcal{I}} = \mathcal{V}(a), \quad a \text{ occurs in } \mathcal{K}, \\ \top^{\mathcal{I}}(s) &= \mathcal{L}(s, \top), \quad \text{for all } s \in \mathbf{S}, \\ \perp^{\mathcal{I}}(s) &= \mathcal{L}(s, \perp), \quad \text{for all } s \in \mathbf{S}, \\ A^{\mathcal{I}}(s) &= \mathcal{L}(s, A), \quad \text{for all } s \in \mathbf{S}, \\ R^{\mathcal{I}}(s, t) &= \mathcal{E}(R, \langle s, t \rangle), \quad \text{for all } \langle s, t \rangle \in \mathbf{S} \times \mathbf{S}. \end{aligned}$$

To prove that $\mathcal{I}$ is a model of $\mathcal{K}$, we show by induction on the structure of concepts that for any $n \in [0, 1]$, if $\mathcal{L}(s, C) \geq n$ then $C^{\mathcal{I}}(s) \geq n$.

(At) If C is $\perp$, $\top$ or a concept name A then the statement is true by construction.

($\neg$) Let $C = \neg A$. If $\mathcal{L}(s, \neg A) \geq n$, then $\mathcal{L}(s, A) \leq 1 - n$. Therefore, $(\neg A)^{\mathcal{I}}(s) = 1 - A^{\mathcal{I}}(s) = 1 - \mathcal{L}(s, A) \geq 1 - (1 - n) = n$.

- ($\sqcap$) Let $C = C_1 \sqcap C_2$. If $\mathcal{L}(s, C_1 \sqcap C_2) \geq n$, from $n \leq \mathcal{L}(s, C_1 \sqcap C_2) \leq \mathcal{L}(s, C_1) \otimes \mathcal{L}(s, C_2)$, there are $n_1, n_2 \in [0, 1]$ such that $\mathcal{L}(s, C_1) \geq n_1$, $\mathcal{L}(s, C_2) \geq n_2$ and $n = n_1 \otimes n_2$. Therefore, by inductive hypothesis on C_1 and C_2 , $n = n_1 \otimes n_2 \leq C_1^{\mathcal{I}}(s) \otimes C_2^{\mathcal{I}}(s) = (C_1 \sqcap C_2)^{\mathcal{I}}(s)$.
- ($\sqcup$) The proof is similar to the case before.
- ($\forall$) Let $C = \forall R.C_1$ and assume $\mathcal{L}(s, \forall R.C_1) \geq n$. Hence, for any $t \in \mathbf{S}$, $\mathcal{L}(t, C_1) \geq \mathcal{E}(R, \langle s, t \rangle) \otimes \mathcal{L}(s, \forall R.C_1)$ and, thus, $\mathcal{L}(t, C_1) \geq R^{\mathcal{I}}(s, t) \otimes n$. By inductive hypothesis on C_1 , for any $t \in \Delta^{\mathcal{I}}$, we have that $C_1^{\mathcal{I}}(t) \geq R^{\mathcal{I}}(s, t) \otimes n$ and, thus, by residuation (Eq. (1)), $R^{\mathcal{I}}(s, t) \Rightarrow C_1^{\mathcal{I}}(t) \geq n$. Therefore, $(\forall R.C_1)^{\mathcal{I}}(s) = \inf_{t \in \mathbf{S}} \{R^{\mathcal{I}}(s, t) \Rightarrow C_1^{\mathcal{I}}(t)\} \geq n$.
- ($\exists$) Let $C = \exists R.C_1$ and assume $\mathcal{L}(s, \exists R.C_1) \geq n$. Therefore, there is $t \in \mathbf{S}$ such that $\mathcal{E}(R, \langle s, t \rangle) \otimes \mathcal{L}(t, C_1) \geq \mathcal{L}(s, \exists R.C_1) \geq n$, i.e. $R^{\mathcal{I}}(s, t) \otimes \mathcal{L}(t, C_1) \geq n$. Let $m' = (R^{\mathcal{I}}(s, t) \otimes \mathcal{L}(t, C_1)) - n$ and define $m = \mathcal{L}(t, C_1) - m'$. Hence, $\mathcal{L}(t, C_1) \geq m$ and $R^{\mathcal{I}}(s, t) \otimes m = n$. Therefore, by inductive hypothesis on C_1 , $n = R^{\mathcal{I}}(s, t) \otimes m \leq R^{\mathcal{I}}(s, t) \otimes C_1^{\mathcal{I}}(t)$ and, thus, $(\exists R.C_1)^{\mathcal{I}}(s) = \sup_{t \in \Delta^{\mathcal{I}}} \{R^{\mathcal{I}}(s, t) \otimes C_1^{\mathcal{I}}(t)\} \geq n$.

Now, as T satisfies Properties 7 and 8 of a fuzzy tableau, this implies that $\mathcal{I}$ is a model of $\mathcal{A}$. Indeed, if $\langle a : C, n \rangle \in \mathcal{A}$, then by definition $\mathcal{L}(\mathcal{V}(a), C) \geq n$ and, thus, $C^{\mathcal{I}}(a^{\mathcal{I}}) \geq n$, while if $\langle (a, b) : R, n \rangle \in \mathcal{A}$, then $\mathcal{E}(R, \langle \mathcal{V}(a), \mathcal{V}(b) \rangle) \geq n$ and, thus, $R^{\mathcal{I}}(a^{\mathcal{I}}, b^{\mathcal{I}}) \geq n$.

As T satisfies Property 9 of a fuzzy tableau, this implies that $\mathcal{I}$ is a model of $\mathcal{T}$. Assume $\langle \top \sqsubseteq C, n \rangle \in \mathcal{T}$. By definition of fuzzy tableau, for all $s \in \mathbf{S}$, $\mathcal{L}(s, C) \geq n$ and, thus, $C^{\mathcal{I}}(s) \geq n$. Since, $(\top \sqsubseteq C)^{\mathcal{I}} = \inf_{s \in \mathbf{S}} \{1 \Rightarrow C^{\mathcal{I}}(s)\} = \inf_{s \in \mathbf{S}} \{C^{\mathcal{I}}(s)\}$, hence $(\top \sqsubseteq C)^{\mathcal{I}} \geq n$ and, thus, $\mathcal{I}$ is a model of $\mathcal{T}$.

Finally, since $\mathbf{S}$ is a finite set, then so is $\Delta^{\mathcal{I}}$ and, therefore, $\mathcal{I}$ is a finite model.

($\Rightarrow$) For the converse, let $\mathcal{I}$ be a finite model of $\mathcal{K}$. Then a fuzzy tableau $T = (\mathbf{S}, \mathcal{L}, \mathcal{E}, \mathcal{V})$ for $\mathcal{K}$ can be defined as follows:

$$\begin{aligned} \mathbf{S} &= \Delta^{\mathcal{I}}, \quad \mathcal{V}(a) = a^{\mathcal{I}} \\ \mathcal{E}(R, \langle s, t \rangle) &= R^{\mathcal{I}}(s, t) \\ \mathcal{L}(s, C) &= C^{\mathcal{I}}(s). \end{aligned}$$

Let us show that T is a fuzzy tableau for $\mathcal{K}.$ (At)

$0 = \perp^{\mathcal{I}}(s) = \mathcal{L}(s, \perp)$. Similarly for $\top$.

($\neg$) By definition, $\mathcal{L}(s, \neg A) = (\neg A)^{\mathcal{I}}(s) = 1 - A^{\mathcal{I}}(s) = 1 - \mathcal{L}(s, A)$.

($\sqcap$) $\mathcal{L}(s, C_1 \sqcap C_2) = (C_1 \sqcap C_2)^{\mathcal{I}}(s) = C_1^{\mathcal{I}}(s) \otimes C_2^{\mathcal{I}}(s) = \mathcal{L}(s, C_1) \otimes \mathcal{L}(s, C_2)$.

($\sqcup$) The proof is similar to the case before.

($\forall$) For any $t \in \mathbf{S}$, $\mathcal{L}(s, \forall R.C) = (\forall R.C)^{\mathcal{I}}(s) \leq R^{\mathcal{I}}(s, t) \Rightarrow C^{\mathcal{I}}(t) = \mathcal{E}(R, \langle s, t \rangle) \Rightarrow \mathcal{L}(t, C)$ and, thus, by residuation (Eq. (1)), $\mathcal{L}(t, C) \geq \mathcal{E}(R, \langle s, t \rangle) \otimes \mathcal{L}(s, \forall R.C)$.

($\exists$) By definition and since $\mathcal{I}$ is witnessed (recall that $\mathcal{I}$ is finite), there is $t \in \mathbf{S}$ such that $\mathcal{L}(s, \exists R.C) = (\exists R.C)^{\mathcal{I}}(s) = R^{\mathcal{I}}(s, t) \otimes C^{\mathcal{I}}(t) = \mathcal{E}(R, \langle s, t \rangle) \otimes \mathcal{L}(t, C)$.

(CAss) Assume $\langle a : C, n \rangle \in \mathcal{A}$. As $\mathcal{I}$ is a model of $\mathcal{A}$, $n \leq C^{\mathcal{I}}(a^{\mathcal{I}}) = \mathcal{L}(\mathcal{V}(a), C)$.

(RAss) Assume $\langle (a, b) : R, n \rangle \in \mathcal{A}$. As $\mathcal{I}$ is a model of $\mathcal{A}$, $n \leq R^{\mathcal{I}}(a^{\mathcal{I}}, b^{\mathcal{I}}) = \mathcal{E}(R, \langle \mathcal{V}(a), \mathcal{V}(b) \rangle)$.

($\sqsubseteq$) Assume $\langle \top \sqsubseteq C, n \rangle \in \mathcal{T}$. As $\mathcal{I}$ is a model of $\mathcal{T}$, $n \leq (\top \sqsubseteq C)^{\mathcal{I}} = \inf_{s \in \Delta^{\mathcal{I}}} \{C^{\mathcal{I}}(s)\} = \inf_{s \in \mathbf{S}} \{\mathcal{L}(s, C)\}$. Therefore, for all $s \in \mathbf{S}$, $\mathcal{L}(s, C) \geq n$.

Therefore, T is a fuzzy tableau for $\mathcal{K}$. Moreover, since $\Delta^{\mathcal{I}}$ is a finite set, then so is $\mathbf{S}$ and, therefore, T is a finite tableau. $\square$

4.2. The case of $\mathcal{L}$ -ALC without TBoxes

In the following, we will assume that the TBox in a knowledge base is empty. Like the tableau algorithm presented for crisp DLs [36,35], and like in [44,45,48,52], our algorithm works on *completion-forests* since an ABox might contain several individuals with arbitrary roles connecting them. Our method combines appropriate DL tableaux rules with methods developed in the context of *many-valued logics* (MVLs) [27].

Let $\mathcal{K}$ be a fuzzy KB with empty TBox. A *completion-forest* $\mathcal{F}$ for $\mathcal{K}$ is a collection of trees whose distinguished roots are arbitrarily connected by edges. Each node v is labelled with a set $\mathcal{L}(v)$ of concepts $C \in \text{sub}(\mathcal{K})$. If $C \in \mathcal{L}(v)$ then we consider a variable $x_{v,C}$. The intuition here is that v is an instance of C to degree equal or greater than the value of the variable $x_{v,C}$. Each edge $\langle v, w \rangle$ is labelled with a set $\mathcal{L}(\langle v, w \rangle)$ of roles $R \in \mathbf{R}_{\mathcal{K}}$ and if $R \in \mathcal{L}(\langle v, w \rangle)$ then we consider a variable $x_{\langle v, w \rangle, R}$ representing the degree of being $\langle v, w \rangle$ and instance of R .

If nodes v and w are connected by an edge $\langle v, w \rangle$ with $R \in \mathcal{L}(\langle v, w \rangle)$ then w is called an *R-successor* of v .

The forest has associated a set $\mathcal{C}_{\mathcal{F}}$ of constraints c of the form $l \leq l'$, $l = l'$, where l, l' are linear expressions. Essentially the constraints are added by the application of rules to concept expressions to reflect the semantics of the connectives. Intuitively, if for a node v , $\mathcal{L}(v)$ contains $C_1 \sqcup C_2$ and $\mathcal{C}_{\mathcal{F}}$ contains $x_{v, C_1 \sqcup C_2} \geq 0.7$ (dictating that v is an instance of $C_1 \sqcup C_2$ to degree at least 0.7), then the application of the ($\sqcup$) rule to $C_1 \sqcup C_2$ will add the concepts C_1 and C_2 to $\mathcal{L}(v)$ and the constraints, encoding $x_{v, C_1} \oplus x_{v, C_2} \geq x_{v, C_1 \sqcup C_2}$, to $\mathcal{C}_{\mathcal{F}}$. Any assignment to the variables that satisfies these constraints, guarantees then also that indeed v is an instance of $C_1 \sqcup C_2$ to degree at least 0.7.

We recall that $n \leq x_1 \oplus n_2$, $n \leq n_1 \otimes n_2$ and $n \geq x_1 \otimes n_2$, where $0 \leq x_i$, $n \leq 1$, can be encoded as the sets of constraints:

$$\begin{aligned} n \leq n_1 \oplus n_2 &\mapsto \{n \leq n_1 + n_2\}, \\ n \leq n_1 \otimes n_2 &\mapsto \{m \leq 1 - n, n_1 + n_2 - 1 \geq n - m, m \in \{0, 1\}\}, \\ n \geq n_1 \otimes n_2 &\mapsto \{n_1 + n_2 - 1 \leq n\}. \end{aligned} \quad (2)$$

Note that only $\otimes$ introduces a new binary variable $m \in \{0, 1\}$. In order to improve the readability of the proofs we show the equivalence between the above expressions and the respective set of constraints once and for all.

Lemma 14. *Let $0 \leq n_i$, $n \leq 1$ and $m \in \{0, 1\}$, then the following hold:*

1. $n \leq n_1 \oplus n_2$ iff the constraint $n \leq n_1 + n_2$ has a solution;
2. $n \leq n_1 \otimes n_2$ iff the set of constraints $m \leq 1 - n$, $n_1 + n_2 - 1 \geq n - m$ has a solution;
3. $n \geq n_1 \otimes n_2$ iff the constraint $n_1 + n_2 - 1 \leq n$ has a solution.

Proof. Let $0 \leq n_i$, $n \leq 1$ and $m \in \{0, 1\}$, then:

1. Suppose that $n \leq n_1 \oplus n_2$, then $n \leq \min\{1, n_1 + n_2\} \leq n_1 + n_2$, hence, the constraint $n \leq n_1 + n_2$ has a solution. Suppose now that $n > n_1 \oplus n_2$. Since $n \leq 1$, then $n_1 \oplus n_2 < 1$ and, therefore, $n_1 + n_2 = n_1 \oplus n_2 < n$. Hence, the constraint $n \leq n_1 + n_2$ has no solution.
2. Suppose that $n \leq n_1 \otimes n_2$, then $n \leq \max\{0, n_1 + n_2 - 1\}$. Either $n_1 + n_2 - 1 > 0$ or $n_1 + n_2 - 1 \leq 0$. In the former case, for $m = 0$, we have that $n - 0 = n \leq \max\{0, n_1 + n_2 - 1\} = n_1 + n_2 - 1$ and the set of constraints has a solution. In the latter case $n = 0$ and for $m = 1$, since $0 \leq n_i$, then $n - m = 0 - 1 \leq n_1 + n_2 - 1$, so, the set of constraints has a solution. Suppose now that $n > n_1 \otimes n_2 = \max\{0, n_1 + n_2 - 1\} \geq 0$. Since $n > 0$, then for $m = 0$, $n - m = n > n_1 + n_2 - 1$, so, the set of constraints has no solution.
3. Suppose that $n \geq n_1 \otimes n_2$, then $n \geq \max\{0, n_1 + n_2 - 1\} \geq n_1 + n_2 - 1$, hence, the constraint $n_1 + n_2 - 1 \leq n$ has a solution. Suppose now that $n < n_1 \otimes n_2$. Since $n \geq 0$, then $n_1 \otimes n_2 > 0$ and, therefore, $n_1 + n_2 - 1 = n_1 \otimes n_2 > n$. Hence, the constraint $n_1 + n_2 - 1 \leq n$ has no solution. $\square$

In what follows, we will be using the expressions $n \leq n_1 \oplus n_2$, $n \leq n_1 \otimes n_2$ and $n \geq n_1 \otimes n_2$ both to denote the inequality with respect to a t -norm (t -conorm) operation and the set of constraints they are equivalent to.

Now, given $\mathcal{K} = \langle \emptyset, \mathcal{A} \rangle$, the algorithm initializes a forest $\mathcal{F}$ as follows:

- $\mathcal{F}$ contains a root node v_0^i , for each individual a_i occurring in $\mathcal{A}$;
- $\mathcal{F}$ contains an edge $\langle v_0^i, v_0^j \rangle$, for each assertion axiom $\langle (a_i, a_j) : R, n \rangle \in \mathcal{A}$;
- for each $\langle a_i : C, n \rangle \in \mathcal{A}$, we add both C to $\mathcal{L}(v_0^i)$ and $x_{v_0^i, C} \geq n$ to $\mathcal{C}_{\mathcal{F}}$;
- for each $\langle (a_i, a_j) : R, n \rangle \in \mathcal{A}$, we add both R to $\mathcal{L}(\langle v_0^i, v_0^j \rangle)$ and $x_{(v_0^i, v_0^j), R} \geq n$ to $\mathcal{C}_{\mathcal{F}}$.

The completion-forest $\mathcal{F}$ is then expanded by repeatedly applying the completion rules described below. The completion-forest is *complete* when none of the completion rules is applicable. Then, the bMILP (bounded Mixed Integer Linear Programming) problem on the set of constraints $\mathcal{C}_{\mathcal{F}}$ is solved. We will show that if there is a solution to $\mathcal{C}_{\mathcal{F}}$ then $\mathcal{K}$ is satisfiable and vice versa.

We also assume a fixed rule application strategy as the order of rules below, such that the rule for $(\exists)$ is applied as last. Also, all expressions in node labels are processed according to the order they are introduced into the completion-forest $\mathcal{F}$.

The rules for $\mathcal{L}$ - $\mathcal{ALC}$ are described below.

- (var). For variable $x_{v, C}$ occurring in $\mathcal{C}_{\mathcal{F}}$ add $x_{v, C} \in [0, 1]$ to $\mathcal{C}_{\mathcal{F}}$. For variable $x_{(v, w): R}$ occurring in $\mathcal{C}_{\mathcal{F}}$ add $x_{(v, w): R} \in [0, 1]$ to $\mathcal{C}_{\mathcal{F}}$.
- ($\perp$). If $\perp \in \mathcal{L}(v)$ then $\mathcal{C}_{\mathcal{F}} := \mathcal{C}_{\mathcal{F}} \cup \{x_{v, \perp} = 0\}$.
- ($\top$). If $\top \in \mathcal{L}(v)$ then $\mathcal{C}_{\mathcal{F}} := \mathcal{C}_{\mathcal{F}} \cup \{x_{v, \top} = 1\}$.
- ($\bar{A}$). If $\bar{A} \in \mathcal{L}(v)$ then add A to $\mathcal{L}(v)$, and $\mathcal{C}_{\mathcal{F}} := \mathcal{C}_{\mathcal{F}} \cup \{x_{v, A} \leq 1 - x_{v, \bar{A}}\}$.
- ($\sqcap$). If (i) $C_1 \sqcap C_2 \in \mathcal{L}(v)$ and (ii) the rule has not been already applied to this concept, then add C_1 and C_2 to $\mathcal{L}(v)$, and $\mathcal{C}_{\mathcal{F}} := \mathcal{C}_{\mathcal{F}} \cup \{x_{v, C_1} \otimes x_{v, C_2} \geq x_{v, C_1 \sqcap C_2}\}$.
- ($\sqcup$). If (i) $C_1 \sqcup C_2 \in \mathcal{L}(v)$ and (ii) the rule has not been already applied to this concept, then add C_1 and C_2 to $\mathcal{L}(v)$, and $\mathcal{C}_{\mathcal{F}} := \mathcal{C}_{\mathcal{F}} \cup \{x_{v, C_1} \oplus x_{v, C_2} \geq x_{v, C_1 \sqcup C_2}\}$.
- ($\forall$). If (i) $\forall R.C \in \mathcal{L}(v)$, $R \in \mathcal{L}(\langle v, w \rangle)$ and (ii) the rule has not been already applied to this concept, then add C to $\mathcal{L}(w)$, and $\mathcal{C}_{\mathcal{F}} := \mathcal{C}_{\mathcal{F}} \cup \{x_{w, C} \geq x_{v, \forall R.C} \otimes x_{(v, w), R}\}$.
- ($\exists$). If (i) $\exists R.C \in \mathcal{L}(v)$ and (ii) the rule has not been already applied to this concept, then create a new node w , add R to $\mathcal{L}(\langle v, w \rangle)$, add C to $\mathcal{L}(w)$, and $\mathcal{C}_{\mathcal{F}} := \mathcal{C}_{\mathcal{F}} \cup \{x_{(v, w), R} \otimes x_{w, C} \geq x_{v, \exists R.C}\}$.

Proposition 15 (Termination). *For each knowledge base $\mathcal{K} = \langle \emptyset, \mathcal{A} \rangle$, the tableau algorithm terminates.*

Proof. Termination is a result of the properties of the expansion rules, as in the classical case [36]. More precisely, we have the following observations. (i) The expansion rules are applied only once for each rule instance, never remove nodes from the tree or concepts from node labels or change the edge labels. (ii) Successors are only generated by the rule $(\exists)$. For any node and for each concept these rules are applied at-most once. (iii) Nodes are labelled with finite non-empty sets of $\text{sub}(\mathcal{K})$ and obviously there is a finite number of possible labelling for a pair of nodes. Therefore, neither there exist infinitely many paths nor there exists an infinitely long path with infinitely many different node labellings. Hence, any path will have a finite length. That is, the algorithm terminates. $\square$

Note that, despite that $|\text{sub}(\mathcal{K})|$ is $O(|\mathcal{K}|)$, we may well have that $|\mathcal{C}_{\mathcal{F}}|$ is $O(2^{|\mathcal{K}|})$, as the completion forest $\mathcal{F}$ may contain exponentially many nodes. Indeed, consider a knowledge base containing only the assertion α

$$\begin{aligned} \alpha : & \exists R.C_1^1 \sqcap \exists R.C_2^1 \sqcap \\ & \forall R.(\exists R.C_1^2 \sqcap \exists R.C_2^2 \sqcap \\ & \forall R.(\exists R.C_1^3 \sqcap \exists R.C_2^3 \sqcap \\ & \dots \\ & \forall R.(\exists R.C_1^n \sqcap \exists R.C_2^n) \dots). \end{aligned}$$

It is not difficult to see that the completion forest $\mathcal{F}$ is a binary tree of depth n consisting of $O(2^n)$ nodes and that $\mathcal{C}_{\mathcal{F}}$ contains both $O(2^n)$ variables taking values in $\{0, 1\}$ as well as $O(2^n)$ variables taking values in $[0, 1]$. As determining whether a MILP problem has a solution is an NP-complete problem [38]⁵ we get an NEXPTIME upper bound for the satisfiability problem (guess the assignment to the binary variables and solve the system of linear in-equations in exponential time).

We next address the soundness of our procedure.

Proposition 16 (Soundness). *If the expansion rules can be applied to a knowledge base $\mathcal{K} = \langle \emptyset, \mathcal{A} \rangle$ such that they yield a complete completion forest $\mathcal{F}$ such that $\mathcal{C}_{\mathcal{F}}$ has a solution, then $\mathcal{K}$ has a finite fuzzy tableau.*

Proof. Let $\mathcal{F}$ be a complete completion-forest constructed by the tableaux algorithm for $\mathcal{K}$. By hypothesis, $\mathcal{C}_{\mathcal{F}}$ has a solution. If l is a linear expression in $\mathcal{C}_{\mathcal{F}}$, with $\bar{l}$ we denote the value of l in this solution.

A fuzzy tableau $T = (\mathbf{S}, \mathcal{L}, \mathcal{E}, \mathcal{V})$ can be defined as follows:

$$\begin{aligned} \mathbf{S} &= \{v \mid v \text{ is a node in } \mathcal{F}\}, \\ \mathcal{V}(a_i) &= v_0^i, \text{ where } v_0^i \text{ is a root node,} \\ \mathcal{L}(v, \perp) &= 0, \text{ for } v \in \mathbf{S}, \\ \mathcal{L}(v, \top) &= 1, \text{ for } v \in \mathbf{S}, \\ \mathcal{L}(v, C) &= \begin{cases} \bar{x}_{v:C}, & \text{if } x_{v:C} \text{ occurs in } \mathcal{C}_{\mathcal{F}}, \\ 0, & \text{otherwise,} \end{cases} \\ \mathcal{E}(R, \langle v, w \rangle) &= \begin{cases} \bar{x}_{(v,w):R}, & \text{if } x_{(v,w):R} \text{ occurs in } \mathcal{C}_{\mathcal{F}}, \\ 0, & \text{otherwise.} \end{cases} \end{aligned}$$

Let us show that that T is a finite funny tableau for $\mathcal{K}$. By construction, as $\mathcal{F}$ is finite, T is finite.

1. By construction, $\mathcal{L}(s, \perp) = 0$ and $\mathcal{L}(s, \top) = 1$, for all $s \in \mathbf{S}$.
2. The $(\bar{A})$ rule ensures that $\bar{x}_{v:A} \leq 1 - \bar{x}_{v:\neg A} \in \mathcal{C}_{\mathcal{F}}$. Therefore, by construction, $\mathcal{L}(v, \neg A) = \bar{x}_{v:\neg A} \leq 1 - \bar{x}_{v:A} = 1 - \mathcal{L}(v, A)$.
3. The $(\sqcap)$ rule ensures that $C_1, C_2 \in \mathcal{L}(v)$ and $x_{v:C_1 \sqcap C_2} \leq x_{v:C_1} \otimes x_{v:C_2} \in \mathcal{C}_{\mathcal{F}}$. Since, by assumption $\mathcal{C}_{\mathcal{F}}$ has a solution, then, by construction $\mathcal{L}(v, C_1 \sqcap C_2) = \bar{x}_{v:C_1 \sqcap C_2} \leq \bar{x}_{v:C_1} \otimes \bar{x}_{v:C_2} = \mathcal{L}(v, C_1) \otimes \mathcal{L}(v, C_2)$.
4. The $(\sqcup)$ rule ensures that $C_1, C_2 \in \mathcal{L}(v)$ and $x_{v:C_1 \sqcup C_2} \leq x_{v:C_1} \oplus x_{v:C_2} \in \mathcal{C}_{\mathcal{F}}$. Since, by assumption $\mathcal{C}_{\mathcal{F}}$ has a solution, then, by construction, $\mathcal{L}(v, C_1 \sqcup C_2) = \bar{x}_{v:C_1 \sqcup C_2} \leq \bar{x}_{v:C_1} \oplus \bar{x}_{v:C_2} = \mathcal{L}(v, C_1) \oplus \mathcal{L}(v, C_2)$.
5. The $(\forall)$ rule ensures that for any node w which is an R -successor of v , $x_{w:C} \geq x_{v:\forall R.C} \otimes x_{(v,w):R} \in \mathcal{C}_{\mathcal{F}}$. Since, by assumption $\mathcal{C}_{\mathcal{F}}$ has a solution, then, by construction $\mathcal{L}(w, C) = \bar{x}_{w:C} \geq \bar{x}_{v:\forall R.C} \otimes \bar{x}_{(v,w):R} = \mathcal{L}(v, \forall R.C) \otimes \mathcal{E}(R, \langle v, w \rangle)$. Therefore, for any $t \in \mathbf{S}$, $\mathcal{L}(t, C) \geq \mathcal{L}(v, \forall R.C) \otimes \mathcal{E}(R, \langle v, t \rangle)$.
6. The $(\exists)$ rule ensures that for a successor w of node v , $x_{v:\exists R.C} \leq x_{(v,w):R} \otimes x_{w:C} \in \mathcal{C}_{\mathcal{F}}$. Since, by assumption $\mathcal{C}_{\mathcal{F}}$ has a solution, then, by construction we have that $\mathcal{L}(v, \exists R.C) = \bar{x}_{v:\exists R.C} \leq \bar{x}_{(v,w):R} \otimes \bar{x}_{w:C} = \mathcal{E}(R, \langle v, w \rangle) \otimes \mathcal{L}(w, C)$. Therefore, there exists a node t such that $\mathcal{L}(v, \exists R.C) \leq \mathcal{E}(R, \langle v, t \rangle) \otimes \mathcal{L}(t, C)$.
7. Assume $\langle a : C, n \rangle \in \mathcal{A}$, then the algorithm initialises a forest $\mathcal{F}$ by adding C to $\mathcal{L}(a)$ and $x_{a:C} \geq n$ to $\mathcal{C}_{\mathcal{F}}$. Hence, by construction $\mathcal{L}(v(a), C) = \bar{x}_{a:C} \geq n$.

⁵ In our case, non-deterministically guess the value of the binary variables y , then check in polynomial time if the system of linear in-equations has a solution for values in $[0, 1]$.

8. Assume $\langle (a, b) : R, n \rangle \in \mathcal{A}$, then the algorithm initialises a forest $\mathcal{F}$ by adding both R to $\mathcal{L}(\langle (a, b) \rangle)$ and $x_{(a,b):R} \geq n$ to $\mathcal{C}_{\mathcal{F}}$ and, thus, by construction $\mathcal{E}(R, \langle \mathcal{V}(a), \mathcal{V}(b) \rangle) = \bar{x}_{(a,b):R} \geq n$.

Therefore, T is a finite fuzzy tableau for $\mathcal{K}$. $\square$

We now prove the completeness of our procedure.

Proposition 17 (Completeness). *If a knowledge base $\mathcal{K} = \langle \emptyset, \mathcal{A} \rangle$ has a finite fuzzy tableau, then the application of the expansion rules yields a complete completion-forest for $\mathcal{K}$ such that $\mathcal{C}_{\mathcal{F}}$ has a solution.*

Proof. Let $T = (\mathbf{S}, \mathcal{L}, \mathcal{E}, \mathcal{V})$ be a finite fuzzy tableau for $\mathcal{K}$. As the tableau algorithm is deterministic, the application of the expansion rules yields a complete completion-forest $\mathcal{F}$ for $\mathcal{K}$. We use the tableau T to inductively show that each rule application preserves the existence of a solution and, thus, there is a solution to $\mathcal{C}_{\mathcal{F}}$. To do this, we will inductively define both a function π , mapping the nodes (individuals) of the forest $\mathcal{F}$ to $\mathbf{S}$ and a variable assignment λ , that assigns to a variable x occurring in $\mathcal{C}_{\mathcal{F}}$ a value in $[0, 1]$ if the constraint $x \in [0, 1]$ is in $\mathcal{C}_{\mathcal{F}}$, while assigns to a variable y occurring in $\mathcal{C}_{\mathcal{F}}$ a value in $\{0, 1\}$ if the constraint $y \in \{0, 1\}$ is in $\mathcal{C}_{\mathcal{F}}$. We will show that indeed λ is a solution for $\mathcal{C}_{\mathcal{F}}$.

At first, we need some auxiliary definitions. In the following, let l be either a constant in $[0, 1]$ or a variable x . With $\lambda(l)$ we indicate the evaluation of the expression l using the variable assignment λ :

$$\lambda(l) = \begin{cases} n, & \text{if } l \text{ is a constant } n \in [0, 1], \\ m, & \text{if } l \text{ is a variable } x \text{ and } \lambda \text{ assigns } m \in [0, 1] \text{ to } x, \\ \text{undefined,} & \text{otherwise.} \end{cases}$$

Next, for $n \in [0, 1]$ and variable x , with $\lambda[x \mapsto n]$ we denote the variable assignment λ' obtained from λ , as follows:

$$\lambda[x \mapsto n] = \begin{cases} \lambda, & \text{if } \lambda(x) \in [0, 1], \text{ i.e. } \lambda(x) \text{ is not undefined,} \\ \lambda \cup \{x \mapsto n\}, & \text{otherwise.} \end{cases}$$

Essentially, $\lambda[x \mapsto n]$ extends λ by assigning n to x if x has not already an assigned value in λ . The extension to multiple parallel assignments is straightforward: for distinct variables $x_1, \dots, x_k$ and values $n_1, \dots, n_k \in [0, 1]$, $\lambda[x_1 \mapsto n_1, \dots, x_k \mapsto n_k]$ is the variable assignment $\lambda[x_1 \mapsto n_1][\dots][x_k \mapsto n_k]$.

We now prove the following claim:

Claim: Let $\mathcal{F}$ be a completion-forest and let π and λ be such that they satisfy condition (*) below. If a rule is applicable to $\mathcal{F}$ then the rule application yields a completion-forest $\mathcal{F}'$ and an extension of π and λ such that (*) is satisfied for $\mathcal{F}'$ as well.

Condition (*) is defined as follows: for any completion forest $\mathcal{F}$, for any node v and w :

$$(*) \begin{cases} \text{if } C \in \mathcal{L}(v) \text{ then } \mathcal{L}(\pi(v), C) \geq \lambda(x_{v:C}), \\ \text{if } w \text{ is an } R\text{-successor of } v \text{ then } \mathcal{E}(R, \langle \pi(v), \pi(w) \rangle) \geq \lambda(x_{(v,w):R}), \\ \lambda \text{ is a solution for } \mathcal{C}_{\mathcal{F}}. \end{cases}$$

Proof of claim: The proof is by induction on rule applications.

Initially, we set $\pi(v_0^i) = \mathcal{V}(a_i)$ for all individuals a_i occurring in $\mathcal{K}$ and let us define λ as assigning to $x_{v_0^i:C}$ the value $\mathcal{L}(\pi(v_0^i), C)$ if $\langle a_i : C, n \rangle \in \mathcal{A}$, while assigning to $x_{(v_0^i, v_0^j):R}$ the value $\mathcal{E}(R, \langle \pi(v_0^i), \pi(v_0^j) \rangle)$ if $\langle (a_i, a_j) : R, m \rangle \in \mathcal{A}$.

Therefore, at the beginning, for $\langle a_i : C, n \rangle \in \mathcal{A}$, we have $C \in \mathcal{L}(v_0^i)$ and $x_{v_0^i:C} \geq n \in \mathcal{C}_{\mathcal{F}}$. As T is a fuzzy tableau for $\mathcal{K}$, T satisfies point 7 of a fuzzy tableau, so $\lambda(x_{v_0^i:C}) = \mathcal{L}(\pi(v_0^i), C) \geq n$. Similarly, for $\langle (a_i, a_j) : R, m \rangle \in \mathcal{A}$, we have $R \in \mathcal{L}(\langle v_0^i, v_0^j \rangle)$ and $x_{(v_0^i, v_0^j):R} \geq m \in \mathcal{C}_{\mathcal{F}}$. As T is a fuzzy tableau for $\mathcal{K}$, T satisfies point 8 of a fuzzy tableau, so $\lambda(x_{(v_0^i, v_0^j):R}) = \mathcal{E}(R, \langle \pi(v_0^i), \pi(v_0^j) \rangle) \geq m$. As a consequence, λ is also a solution for $\mathcal{C}_{\mathcal{F}}$ and, thus, for the initial forest, condition (*) is satisfied.

Now, assume that for a completion-forest $\mathcal{F}$, π and λ satisfy (*). Let us show that after a rule application, we get a completion-forest $\mathcal{F}'$ and an extension of π and λ such that (*) is satisfied for $\mathcal{F}'$ as well.

($\perp$) Assume that ($\perp$) rule has been applied to $\perp \in \mathcal{L}(v)$, then:

1. $\mathcal{F}'$ is obtained from $\mathcal{F}$ by setting $\mathcal{C}_{\mathcal{F}'} := \mathcal{C}_{\mathcal{F}} \cup \{x_{v:\perp} = 0\}$.
2. As (*) is satisfied for $\mathcal{F}$, then $\mathcal{L}(\pi(v), \perp) \geq \lambda(x_{v:\perp}) \geq 0$ and λ is a solution to $\mathcal{C}_{\mathcal{F}}$. As T is a fuzzy tableau, by point 1 of a fuzzy tableau, $\mathcal{L}(\pi(v), \perp) = 0$ and, thus, $\lambda(x_{v:\perp}) = 0$.
3. Therefore, λ is a solution to $\mathcal{C}_{\mathcal{F}'}$ and, thus, (*) is true also for $\mathcal{F}'$, π and λ .

($\top$) The proof is similar to the case before.

($\bar{A}$) Assume that the ($\bar{A}$) rule has been applied to $\neg A \in \mathcal{L}(v)$. Let $l = x_{v:\neg A}$, then:

1. $\mathcal{F}'$ is obtained from $\mathcal{F}$ by (i) adding A to $\mathcal{L}(v)$; and (ii) setting $\mathcal{C}_{\mathcal{F}'} := \mathcal{C}_{\mathcal{F}} \cup \{x_{v:A} \leq 1 - l\}$.

2. as $(*)$ is satisfied for $\mathcal{F}$, then $\mathcal{L}(\pi(v), \neg A) \geq \lambda(l)$ and λ is a solution to $\mathcal{C}_{\mathcal{F}}$. As T is a fuzzy tableau, by point 2, there is n such that $0 \leq n \leq \mathcal{L}(\pi(v), A) \leq 1 - \mathcal{L}(\pi(v), \neg A) \leq 1 - \lambda(l)$.
3. Now, let $\lambda' := \lambda[x_{v:A} \mapsto n]$. Therefore, for $\mathcal{F}'$, (i) $\mathcal{L}(\pi(v), \neg A) \geq \lambda(l) = \lambda'(l)$; (ii) $\mathcal{L}(\pi(v), A) \geq n = \lambda'(x_{v:A})$; and (iii) λ' is a solution to $\mathcal{C}_{\mathcal{F}'}$ and, thus, $(*)$ is true also for $\mathcal{F}'$, π and λ' .

($\cap$) Assume that the ($\cap$) rule has been applied to $C_1 \cap C_2 \in \mathcal{L}(v)$. Let $l = x_{v:C_1 \cap C_2}$, $l_1 = x_{v:C_1}$, and $l_2 = x_{v:C_2}$. Then

1. $\mathcal{F}'$ is obtained from $\mathcal{F}$ by (i) adding C_i to $\mathcal{L}(v)$ ($i = 1, 2$); and (ii) setting $\mathcal{C}_{\mathcal{F}'} := \mathcal{C}_{\mathcal{F}} \cup \{l_1 \otimes l_2 \geq l\}$.
2. as $(*)$ is satisfied for $\mathcal{F}$, then $\mathcal{L}(\pi(v), C_1 \cap C_2) \geq \lambda(l)$ and λ is a solution to $\mathcal{C}_{\mathcal{F}}$. As T is a fuzzy tableau, by point 3. $\mathcal{L}(\pi(v), C_1) \otimes \mathcal{L}(\pi(v), C_2) \geq \mathcal{L}(\pi(v), C_1 \cap C_2) \geq \lambda(l)$. Therefore, there are $n_1, n_2 \in [0, 1]$, such that $\mathcal{L}(\pi(v), C_i) \geq n_i$, and $n_1 \otimes n_2 \geq \lambda(l)$.
3. Now, let $\lambda' := \lambda[l_i \mapsto n_i]$. Therefore, for $\mathcal{F}'$, (i) $\mathcal{L}(\pi(v), C_1 \cap C_2) \geq \lambda(l) = \lambda'(l)$; (ii) $\mathcal{L}(\pi(v), C_i) \geq n_i = \lambda'(l_i)$; and (iii) λ' is a solution to $\mathcal{C}_{\mathcal{F}'}$. Therefore, $(*)$ is true also for $\mathcal{F}'$, π and λ' .

($\sqcup$) The proof is similar to the case before.

($\forall$) Assume that the ($\forall$) rule has been applied to $\forall R.C \in \mathcal{L}(v)$ and node w that is an R -successor of v . Let $l_1 = x_{v:\forall R.C}$, $l_2 = x_{(v,w):R}$ and $l_3 = x_{w:C}$. Then:

1. $\mathcal{F}'$ is obtained from $\mathcal{F}$ by (i) adding C to $\mathcal{L}(w)$; and (ii) setting $\mathcal{C}_{\mathcal{F}'} := \mathcal{C}_{\mathcal{F}} \cup \{l_3 \geq l_1 \otimes l_2\}$.
2. as $(*)$ is satisfied for $\mathcal{F}$, so $\mathcal{L}(\pi(v), \forall R.C) \geq \lambda(l_1)$, $\mathcal{E}(R, \langle \pi(v), \pi(w) \rangle) \geq \lambda(l_2)$ and λ is a solution to $\mathcal{C}_{\mathcal{F}}$. As T is a fuzzy tableau, for node w , $\mathcal{L}(w, C) \geq \mathcal{E}(R, \langle \pi(v), \pi(w) \rangle) \otimes \mathcal{L}(\pi(v), \forall R.C) \geq \lambda(l_2) \otimes \lambda(l_1)$. Therefore, there is $n \in [0, 1]$ such that $\mathcal{L}(\pi(w), C) \geq n \geq \lambda(l_2) \otimes \lambda(l_1)$.
3. Now, let $\lambda' := \lambda[l_i \mapsto n]$. Therefore, for $\mathcal{F}'$, (i) $\mathcal{L}(\pi(v), \forall R.C) \geq \lambda(l_1) = \lambda'(l_1)$; (ii) $\mathcal{E}(R, \langle \pi(v), \pi(w) \rangle) \geq \lambda(l_2) = \lambda'(l_2)$; (iii) $\mathcal{L}(\pi(w), C) \geq n = \lambda'(l_3)$; and (iv) λ' is a solution to $\mathcal{C}_{\mathcal{F}'}$. Therefore, $(*)$ is true also for $\mathcal{F}'$, π and λ' .

($\exists$) Assume that the ($\exists$) rule has been applied to $\exists R.C \in \mathcal{L}(v)$. Then

1. $\mathcal{F}'$ is obtained from $\mathcal{F}$ by (i) creating a new node w ; (ii) adding R to $\mathcal{L}(\langle v, w \rangle)$ and C to $\mathcal{L}(w)$; and (iii) setting $\mathcal{C}_{\mathcal{F}'} := \mathcal{C}_{\mathcal{F}} \cup \{l_1 \otimes l_2 \geq l\}$, where $l = x_{v:\exists R.C}$, $l_1 = x_{(v,w):R}$ and $l_2 = x_{w:C}$.
2. as $(*)$ is satisfied for $\mathcal{F}$, then $\mathcal{L}(\pi(v), \exists R.C) \geq \lambda(l)$ and λ is a solution to $\mathcal{C}_{\mathcal{F}}$. As T is a fuzzy tableau, there is a node $t \in \mathbf{S}$ such that $\mathcal{E}(R, \langle \pi(v), t \rangle) \otimes \mathcal{L}(t, C) \geq \mathcal{L}(\pi(v), \exists R.C) \geq \lambda(l)$. Hence, there are $n_1, n_2 \in [0, 1]$ such that $\mathcal{E}(R, \langle \pi(v), t \rangle) \geq n_1$, $\mathcal{L}(t, C) \geq n_2$ and $n_1 \otimes n_2 \geq \lambda(l)$.
3. Now, let $\pi' := \pi[w \mapsto t]$, $\lambda' := \lambda[l_i \mapsto n_i]$. Therefore, for $\mathcal{F}'$, (i) $\mathcal{L}(\pi'(v), \exists R.C) = \mathcal{L}(\pi(v), \exists R.C) \geq \lambda(l) = \lambda'(l)$; (ii) $\mathcal{E}(R, \langle \pi'(v), \pi'(w) \rangle) = \mathcal{E}(R, \langle \pi(v), t \rangle) \geq n_1 = \lambda'(l_1)$; (iii) $\mathcal{L}(\pi'(w), C) = \mathcal{L}(t, C) \geq n_2 = \lambda'(l_2)$; and (iv) λ' is a solution to $\mathcal{C}_{\mathcal{F}'}$. Therefore, $(*)$ is true also for $\mathcal{F}'$, π' and λ' .

This concludes the proof of the claim.

Now, let T be a finite fuzzy tableau. Then the claim and the fact that the tableau algorithm terminates, guarantee that we can build a complete completion-forest $\mathcal{F}$ for which $\mathcal{C}_{\mathcal{F}}$ has a solution (λ), which concludes the proof of Proposition 17. $\square$

4.3. The case of $\mathcal{L}$ -ALC with acyclic TBoxes

In the following, we show that we may extend the results of the previous section to acyclic TBoxes. Let us consider, thus, an acyclic knowledge base $\mathcal{K} = \langle \mathcal{T}, \mathcal{A} \rangle$. Recall that by Propositions 11 and 13 we may restrict our attention to finite models and finite fuzzy tableaux only.

We next show how to extend the inference rule set of Section 4.2 to deal also with the GCIs of acyclic knowledge bases and then show termination, soundness and completeness of the new calculus. Note that, unlike [8], in which an acyclic TBox is eliminated by systematically replacing any defined atom by means of its definition (this procedure is called *unfolding*), we instead present here an alternative method, called *lazy unfolding* in the classical case [3,4,33] for which it has been shown to be more efficient.⁶

Specifically, let us consider the following rule:

($\sqsubseteq_R$). Assume (i) $\langle \top \sqsubseteq D, n \rangle \in \mathcal{T}$, (ii) D is of the form $A \rightarrow C$ or $A \leftrightarrow C$, (iii) v is a node to which this rule has not yet been applied and (iv) $A \in \mathcal{L}(v)$. Then add D to $\mathcal{L}(v)$ and set $\mathcal{C}_{\mathcal{F}} := \mathcal{C}_{\mathcal{F}} \cup \{x_{v:D} \geq n\}$.

Remark 2. Consider a complete completion forest $\mathcal{F}$ and D of the form $A \rightarrow C$ or $A \leftrightarrow C$. Then obviously, if $A \in \mathcal{L}(v)$ then $\{C, D\} \subseteq \mathcal{L}(v)$ (apply ($\sqsubseteq_R$) to $A \in \mathcal{L}(v)$, then the rules to $D \in \mathcal{L}(v)$). On the other hand, if $\{C, D\} \subseteq \mathcal{L}(v)$ then $A \in \mathcal{L}(v)$ (by applying the rules to D , A and C will be added to $\mathcal{L}(v)$).

Proposition 18 (Termination). For each acyclic knowledge base $\mathcal{K} = \langle \mathcal{T}, \mathcal{A} \rangle$, the tableau algorithm terminates.

⁶ Whether lazy unfolding will be more efficient than unfolding in the fuzzy case as well will be the subject of future investigations, in combination with other optimisation methods inspired by those developed for classical DLs.

Proof. The proof is as for Proposition 15 by additionally taking into account that a completion forest $\mathcal{F}$ for $\mathcal{K}$ is finite as no infinite number of nodes can be created via the GCIs. In fact, $(\sqsubseteq_R)$ does not create a new node, for a node v in $\mathcal{F}$, $\mathcal{L}(v)$ is finite and bounded, as $\text{sub}(\mathcal{K})$ and the set of concept names in $\mathcal{K}$ are finite. Hence, v has only finitely many successors. Furthermore, there cannot be an infinite path in $\mathcal{F}$, as otherwise there is a concept name A and an infinite number of nodes v along that path with $A \in \mathcal{L}(v)$. That is, A uses A in $\mathcal{T}$ and, thus, $\mathcal{K}$ is not acyclic. Therefore, $\mathcal{F}$ has to be finite and, thus, the procedure terminates. $\square$

We next move to prove the soundness. The proof extends the one in Proposition 16. The critical point is the construction of the fuzzy tableau from a solution to the constraint system $\mathcal{C}_{\mathcal{F}}$.

Proposition 19 (Soundness). *If the expansion rules can be applied to an acyclic knowledge base $\mathcal{K} = \langle \mathcal{T}, \mathcal{A} \rangle$ such that they yield a complete completion forest $\mathcal{F}$ such that $\mathcal{C}_{\mathcal{F}}$ has a solution, then $\mathcal{K}$ has a finite fuzzy tableau.*

Proof. The proof is similar to Proposition 16. We have additionally to manage the case involving the GCIs in $\mathcal{T}$. The tableau is constructed as follows:

$$\begin{aligned} \mathbf{S} &= \{v \mid v \text{ is a node in } \mathcal{F}\}, \\ \mathcal{V}(a_i) &= v_0^i, \text{ where } v_0^i \text{ is a root node,} \\ \mathcal{L}(v, \perp) &= 0, \text{ for } v \in \mathbf{S}, \\ \mathcal{L}(v, \top) &= 1, \text{ for } v \in \mathbf{S}. \end{aligned}$$

For any atomic concept A not occurring in the head of a definitional GCI we have

$$\mathcal{L}(v, A) = \begin{cases} \bar{x}_{v,A}, & \text{if } A \in \mathcal{L}(v), \\ 0, & \text{otherwise.} \end{cases}$$

For any role R we have

$$\mathcal{E}(R, \langle v, w \rangle) = \begin{cases} \bar{x}_{\langle v, w \rangle, R}, & \text{if } R \in \mathcal{L}(\langle v, w \rangle), \\ 0, & \text{otherwise.} \end{cases}$$

For any atomic concept A occurring in the head of a definitional GCI, we proceed as follows.⁷ Fix an arbitrary linearisation $\langle A_1, A_2, \dots, A_k \rangle$ of the “uses” partial order on the atomic concepts appearing in the head of definitional GCIs in $\mathcal{T}$ such that if A_i uses A_j then $j < i$. Let

$$\begin{aligned} &\langle \top \sqsubseteq A_k \leftrightarrow C_k, n_k \rangle \\ &\vdots \\ &\langle \top \sqsubseteq A_2 \leftrightarrow C_2, n_2 \rangle \\ &\langle \top \sqsubseteq A_1 \leftrightarrow C_1, n_1 \rangle \end{aligned}$$

Be these definitional GCIs in $\mathcal{T}$. Note that if $1 \leq j \leq i \leq k$ then A_i cannot occur in C_j . Note also that if $A_i \in \mathcal{L}(v)$ then $C_i \in \mathcal{L}(v)$ (by Remark 2) and, thus, if $C_i \notin \mathcal{L}(v)$ then $A_i \notin \mathcal{L}(v)$.

Now, we extend $\mathcal{L}$ incrementally to the atomic concepts A_i as follows. For $i = 1 \dots k$

$$\mathcal{L}(v, A_i) = \begin{cases} \bar{x}_{v,A_i}, & \text{if } A_i \in \mathcal{L}(v), \\ \bar{x}_{v,C_i}, & \text{if } A_i \notin \mathcal{L}(v) \text{ and } C_i \in \mathcal{L}(v), \\ \mathcal{L}^*(v, C_i), & \text{otherwise i.e., } C_i \notin \mathcal{L}(v), \end{cases}$$

where $\mathcal{L}^*(v, C_i)$ is the structural extension of $\mathcal{L}$ to C_i . The structural extension is defined inductively on the structure of concepts D as follows:

$$\mathcal{L}^*(v, D) = \begin{cases} \mathcal{L}(v, A), & \text{if } D \text{ is atomic concept } A, \\ 1 - \mathcal{L}(v, A), & \text{if } D \text{ is } \neg A, \\ \mathcal{L}^*(v, C_1) \otimes \mathcal{L}^*(v, C_2), & \text{if } D \text{ is } C_1 \sqcap C_2, \\ \mathcal{L}^*(v, C_1) \oplus \mathcal{L}^*(v, C_2), & \text{if } D \text{ is } C_1 \sqcup C_2, \\ \min_{w \in \mathbf{S}} \mathcal{E}(R, \langle v, w \rangle) \Rightarrow \mathcal{L}^*(w, C), & \text{if } D \text{ is } \forall R.C, \\ \max_{w \in \mathbf{S}} \mathcal{E}(R, \langle v, w \rangle) \otimes \mathcal{L}^*(w, C), & \text{if } D \text{ is } \exists R.C. \end{cases}$$

Note that $\mathcal{L}^*(v, C_i)$ is well defined as, by construction, all atomic concepts and roles occurring in C_i have an $\mathcal{L}$ value.

⁷ The method is inspired by [37], Theorem 3.4.

Eventually, for any non-atomic concept D , we set⁸

$$\mathcal{L}(v, D) = \begin{cases} \bar{x}_{v:D}, & \text{if } D \in \mathcal{L}(v), \\ \mathcal{L}^*(w, D), & \text{otherwise.} \end{cases}$$

It remains to show that T is a finite fuzzy tableau for $\mathcal{K}$ for the GCI case. Specifically, w.r.t. [Proposition 16](#), it suffices to show that $\mathcal{L}$ satisfies the conditions in [Definition 12](#) for a concept D of the form $A \rightarrow C$ or $A \leftrightarrow C$. Assume $\langle \top \sqsubseteq D, n \rangle \in \mathcal{T}$ and consider a node v . We have to show that

- $\mathcal{L}(v, D) \geq n$; and
- $\mathcal{L}(v, D) \leq \mathcal{L}(v, A) \bullet \mathcal{L}(v, C)$, where $\bullet \in \{\Rightarrow, \Leftarrow\}$.

Case D is $A \rightarrow C$.

If $D \in \mathcal{L}(v)$ then, by [Remark 2](#), $\{A, C, D\} \subseteq \mathcal{L}(v)$ and, thus, by construction of the fuzzy tableau and of $\mathcal{C}_{\mathcal{F}}$, we have that $\mathcal{L}(v, D) = \bar{x}_{v:D} \geq n$ and

$$\mathcal{L}(v, A) \Rightarrow \mathcal{L}(v, C) = \bar{x}_{v:A} \Rightarrow \mathcal{L}(v, C) = \bar{x}_{v:A} \Rightarrow \bar{x}_{v:C} \geq \bar{x}_{v:A \rightarrow C} = \bar{x}_{v:D} = \mathcal{L}(v, D).$$

If $D \notin \mathcal{L}(v)$ then, by [Remark 2](#), $A \notin \mathcal{L}(v)$. Hence, by construction $\mathcal{L}(v, A) = 0$ and, therefore,

$$\mathcal{L}(v, D) = \mathcal{L}^*(v, D) = \mathcal{L}^*(v, A \rightarrow C) = \mathcal{L}^*(v, A) \Rightarrow \mathcal{L}^*(v, C) = \mathcal{L}(v, A) \Rightarrow \mathcal{L}^*(v, C) = 0 \Rightarrow \mathcal{L}^*(v, C) = 1 \geq n.$$

Moreover,

$$\mathcal{L}(v, A) \Rightarrow \mathcal{L}(v, C) = 0 \Rightarrow \mathcal{L}(v, C) = 1 \geq \mathcal{L}(v, D).$$

Case D is $A \leftrightarrow C$.

If $D \in \mathcal{L}(v)$ then, by [Remark 2](#), $\{A, C, D\} \subseteq \mathcal{L}(v)$ and, thus, by construction of the fuzzy tableau and of $\mathcal{C}_{\mathcal{F}}$, we have that $\mathcal{L}(v, D) = \bar{x}_{v:D} \geq n$ and

$$\mathcal{L}(v, A) \Leftarrow \mathcal{L}(v, C) = \bar{x}_{v:A} \Leftarrow \bar{x}_{v:C} \geq \bar{x}_{v:A \leftrightarrow C} = \bar{x}_{v:D} = \mathcal{L}(v, D).$$

If $D \notin \mathcal{L}(v)$ then, by [Remark 2](#), $A \notin \mathcal{L}(v)$. We have now two cases, either $C \in \mathcal{L}(v)$ or $C \notin \mathcal{L}(v)$. If $C \in \mathcal{L}(v)$ then by construction, $\mathcal{L}(v, A) = \bar{x}_{v:C} = \mathcal{L}(v, C)$. Therefore,

$$\mathcal{L}(v, D) = \mathcal{L}^*(v, D) = \mathcal{L}^*(v, A \leftrightarrow C) = \mathcal{L}^*(v, A) \Leftarrow \mathcal{L}^*(v, C) = \mathcal{L}(v, A) \Leftarrow \mathcal{L}(v, C) = \bar{x}_{v:C} \Leftarrow \bar{x}_{v:C} = 1 \geq n.$$

If $C \notin \mathcal{L}(v)$ then by construction, $\mathcal{L}(v, A) = \mathcal{L}^*(v, C) = \mathcal{L}(v, C)$. Therefore,

$$\mathcal{L}(v, D) = \mathcal{L}^*(v, D) = \mathcal{L}^*(v, A \leftrightarrow C) = \mathcal{L}^*(v, A) \Leftarrow \mathcal{L}^*(v, C) = \mathcal{L}(v, A) \Leftarrow \mathcal{L}(v, C) = \mathcal{L}(v, A) \Leftarrow \mathcal{L}(v, A) = 1 \geq n.$$

Therefore, all conditions of a fuzzy tableau are satisfied, which completes the proof. $\square$

Proposition 20 (Completeness). *If an acyclic knowledge base $\mathcal{K} = \langle \mathcal{T}, \mathcal{A} \rangle$ has a finite fuzzy tableau, then the application of the expansion rules yields a complete completion-forest for $\mathcal{K}$ such that $\mathcal{C}_{\mathcal{F}}$ has a solution.*

Proof. The proof is similar as for [Proposition 17](#), where we have to add the GCI case in the proof of the claim:

($\sqsubseteq_R$)

assume that the ($\sqsubseteq_R$) has been applied to axiom $\langle \top \sqsubseteq D, n \rangle$ on node v . Therefore, D is of the form $A \rightarrow C$ or $A \leftrightarrow C$ and $A \in \mathcal{L}(v)$. Let $l = x_{v:D}$, then:

1. $\mathcal{F}'$ is obtained from $\mathcal{F}$ by adding D to $\mathcal{L}(v)$ and by setting $\mathcal{C}_{\mathcal{F}'} := \mathcal{C}_{\mathcal{F}} \cup \{l \geq n\}$.
2. As $(*)$ is satisfied for $\mathcal{F}$, λ is a solution to $\mathcal{C}_{\mathcal{F}}$. As T is a fuzzy tableau, by point 9, for all $s \in \mathbf{S}$, $\mathcal{L}(s, D) \geq n$ and, thus, $\mathcal{L}(\pi(v), D) \geq n$. Therefore, there is $m \in [0, 1]$ such that $\mathcal{L}(\pi(v), D) \geq m \geq n$.
3. Now, let $\lambda' := \lambda[l \mapsto m]$. Therefore, for $\mathcal{F}'$, (i) $\mathcal{L}(\pi(v), D) \geq m = \lambda'(l)$; and (ii) λ' is a solution to $\mathcal{C}_{\mathcal{F}'}$. Therefore, $(*)$ is true also for $\mathcal{F}'$, π and λ' .

This completes the proof. $\square$

⁸ Note that $\mathcal{L}^*(w, D)$ is well defined as now $\mathcal{L}(v, A)$ is defined for all atomic concepts A and nodes v , due to the linearisation of the uses relation.

Therefore, by relying on Remark 1, we have immediately

Corollary 21. *For $\mathbb{L}$ -ALC with acyclic TBoxes, there is a MILP based decision procedure for the knowledge base satisfiability problem, the best entailment degree problem and the best satisfiability degree problem.*

4.4. The case of $\mathbb{L}_n$ -ALC

For the sake of completeness, we recall here also some known results dealing with the case in which the truth space is finite and of the form $(n \geq 3)$.

$$L_n = \left\{ 0, \frac{1}{n-1}, \dots, \frac{n-2}{n-1}, 1 \right\}.$$

We denote the n -valued language as $\mathbb{L}_n$ -ALC. Concerning the knowledge base satisfiability problem, its decidability is already known [12,18]. In fact, by adapting the method described in [47], which is based on a reduction of a finitely-valued DL to a classical DL, [18] shows the decidability of the KB satisfiability problem for $\mathbb{L}_n$ -SROIQ [34], a significantly more expressive language than $\mathbb{L}_n$ -ALC. A related result is [22] that shows that the formula m -satisfiability problem⁹ in the minimal Łukasiewicz Multi-modal Logic is PSPACE-complete. Since it can be easily proven that this problem is equivalent to the concept m -satisfiability w.r.t. an empty TBox, we have that this problem is PSPACE-complete for $\mathbb{L}_n$ -ALC. Moreover, [19] shows that deciding concept satisfiability w.r.t. a TBox is an EXPTIME-complete problem for $\mathbb{L}_n$ -ALC. Similarly, [20] shows that for ALC over finite residuated lattices, the problems of satisfiability and subsumption of concepts in this logic are EXPTIME-complete w.r.t. a TBox and PSPACE-complete w.r.t. an acyclic TBox (see also [49] for analogous results).

5. Conclusions and outlook

We have presented undecidability results related to reasoning problems for fuzzy DLs with GCIs under Łukasiewicz Logic and provided a formal proof of a MILP based reasoning procedure for the special case of acyclic TBoxes. As a side product, some results may have an impact on other logics such as

- *Hybrid logics* [2], which are roughly modal logics allowing to talk about the worlds. This is not surprising due to the close relationship between DLs and Propositional Multimodal Logic **K** [43] (intuitively, $\forall R.C \mapsto \Box_R C$ and $\exists R.C \mapsto \Diamond_R C$ and $a:C \mapsto @_a C$ and $\top \sqsubseteq C \mapsto \Box_U C$, where U is an universal accessibility relation¹⁰). Specifically we conjecture that the satisfiability problem for propositional fuzzy multimodal hybrid logic allowing an universal modality is undecidable under Łukasiewicz Logic, even if the accessibility relation is crisp.
- *Guarded fragments of first-order logics* [1,25,26], where roughly formulae are of the form $\exists \vec{y}. \alpha(\mathbf{x}, \mathbf{y}) \wedge \varphi(\mathbf{x}, \mathbf{y})$ and $\forall \vec{y}. \alpha(\mathbf{x}, \mathbf{y}) \rightarrow \varphi(\mathbf{x}, \mathbf{y})$, where $\alpha(\mathbf{x}, \mathbf{y})$ is atomic. The correspondence to DLs is evident in light of the fact that $(\exists R.C)(x) \mapsto \exists y. R(-, x, y) \wedge C(y)$ and $(\forall R.C)(x) \mapsto \forall y. R(x, y) \rightarrow C(y)$. We conjecture that the satisfiability problem for an appropriate guarded fragment of fuzzy first-order logic is undecidable under Łukasiewicz Logic, even if the guards are crisp.

For future work, we envisage several interesting issues to be addressed:

1. A systematic investigation about the computational complexity of reasoning within fuzzy DLs with respect to several parameters such as:
 - (a) the fuzzy DL language at hand;
 - (b) the truth space (e.g., a priori fixed and finite, finite but arbitrary of size, infinite, other algebraic structures);
 - (c) the t -norm or some other finite function defined via matrixes [28].
2. The development of efficient reasoning procedures that scale.

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⁹ For a given modal formula φ and $m \in [0, 1]$, determine whether there is an $\mathbb{L}_n$ -valued Kripke frame that gives value m to φ . In terms of FDL this problem can be translated as the problem of determining, for a given concept C and $m \in [0, 1]$, whether there is an $\mathcal{I}$ and $x \in \mathcal{A}^{\mathcal{I}}$ such that $C^{\mathcal{I}}(x) = m$.

¹⁰ For all worlds w, w' , $U(w, w')$ holds.

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